

STUDENT NUMBER: \_\_\_\_\_



*Pymble Ladies' College*

## 2023 Chemistry Trial Examination

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### General Instructions

- Reading time – 5 minutes
- Working time – 3 hours
- Write using black pen
- Draw diagrams using pencil
- NESA approved calculators may be used
- A formulae sheet, data sheet and Periodic Table are provided separately
- For questions in Section I, record your responses on the multiple choice answer sheet provided at the back of this paper.
- For questions in Section II, show all relevant working in questions involving calculations.

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### Total Marks: 100

#### Section I – 20 marks

- Attempt Questions 1 – 20
- Allow about 35 minutes for this section

#### Section II – 80 marks

- Attempt Questions 21 – 32
- Allow about 2 hours and 25 minutes for this section

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## Section I

20 marks

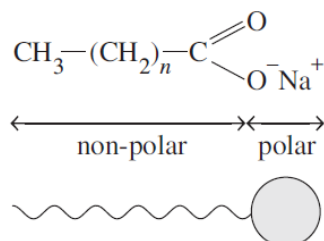
Attempt Questions 1 – 20

Allow about 35 minutes for this section

Use the multiple-choice answer sheet for Questions 1 – 20.

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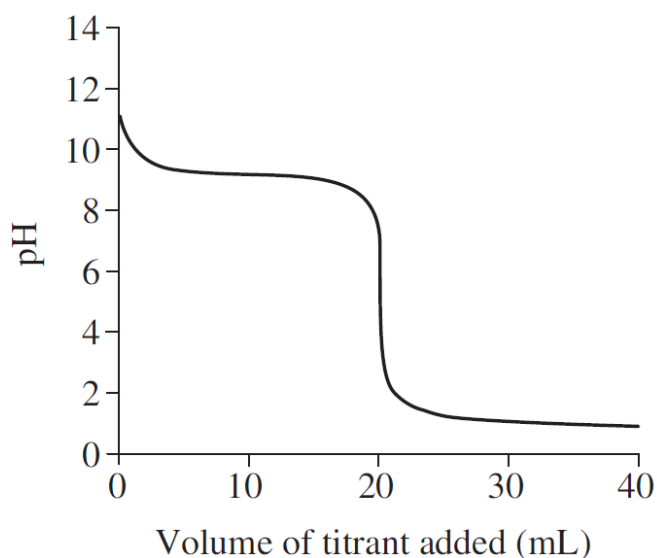
1. Consider the diagram.



What structure does this diagram represents?

- A. soap molecule.
  - B. hydrocarbon.
  - C. cationic detergent molecule.
  - D. anionic detergent molecule.
2. What is the conjugate base of  $\text{HSO}_4^-$ ?
- A.  $\text{H}^+$
  - B.  $\text{SO}_4^{2-}$
  - C.  $\text{H}_2\text{SO}_4$
  - D.  $\text{SO}_4^{2-}$
3. Which option is the correct equation for the incomplete combustion of ethanol?
- A.  $\text{C}_2\text{H}_5\text{OH}(\text{l}) + \frac{1}{2} \text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g}) + 3\text{H}_2(\text{g})$
  - B.  $\text{C}_2\text{H}_5\text{OH}(\text{l}) + \frac{3}{2} \text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2(\text{g})$
  - C.  $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 2\text{O}_2(\text{g}) \rightarrow 2\text{CO}(\text{g}) + 3\text{H}_2\text{O}(\text{l})$
  - D.  $\text{C}_2\text{H}_5\text{OH}(\text{l}) + 3\text{O}_2(\text{g}) \rightarrow 2\text{CO}_2(\text{g}) + 3\text{H}_2\text{O}(\text{l})$
4. Which of the following statements about activation energy,  $E_a$ , and equilibrium is correct?
- A. Lowering the  $E_a$  will cause a reaction to reach equilibrium faster.
  - B. Lowering the  $E_a$  will shift the position of equilibrium to increase product yield.
  - C.  $E_a$  affects only exothermic equilibrium reactions.
  - D.  $E_a$  does not apply to equilibrium reactions.

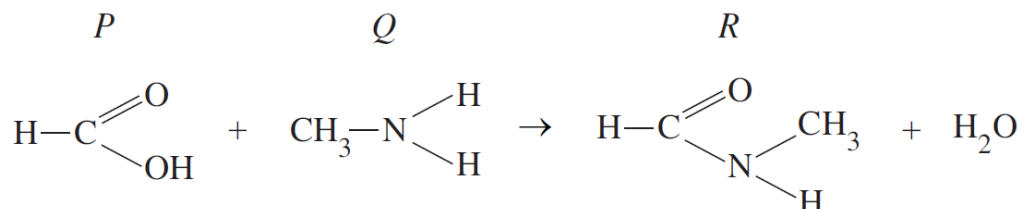
5. Consider the titration curve.



Based on the curve, what can be concluded about the titration?

- A. A strong acid was added to a strong base.
- B. A strong acid was added to a weak base.
- C. A strong base was added to a weak acid.
- D. A weak acid was added to a weak base.

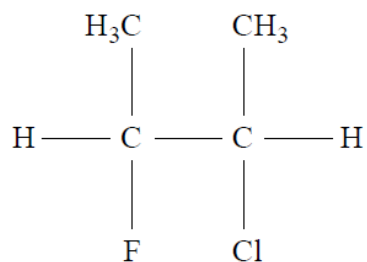
6. The diagram shows molecules *P* and *Q* reacting to form molecule *R* and water.



Which row of the table identifies the molecule types?

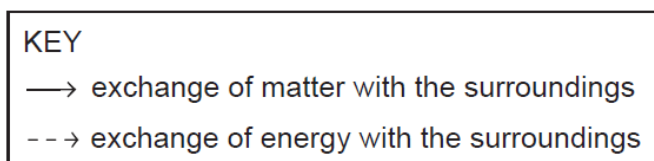
|    | <i>P</i>        | <i>Q</i> | <i>R</i> |
|----|-----------------|----------|----------|
| A. | alcohol         | amine    | ester    |
| B. | carboxylic acid | amide    | amine    |
| C. | carboxylic acid | amine    | amide    |
| D. | aldehyde        | amide    | ketone   |

7. What is the IUPAC name of the molecule below?

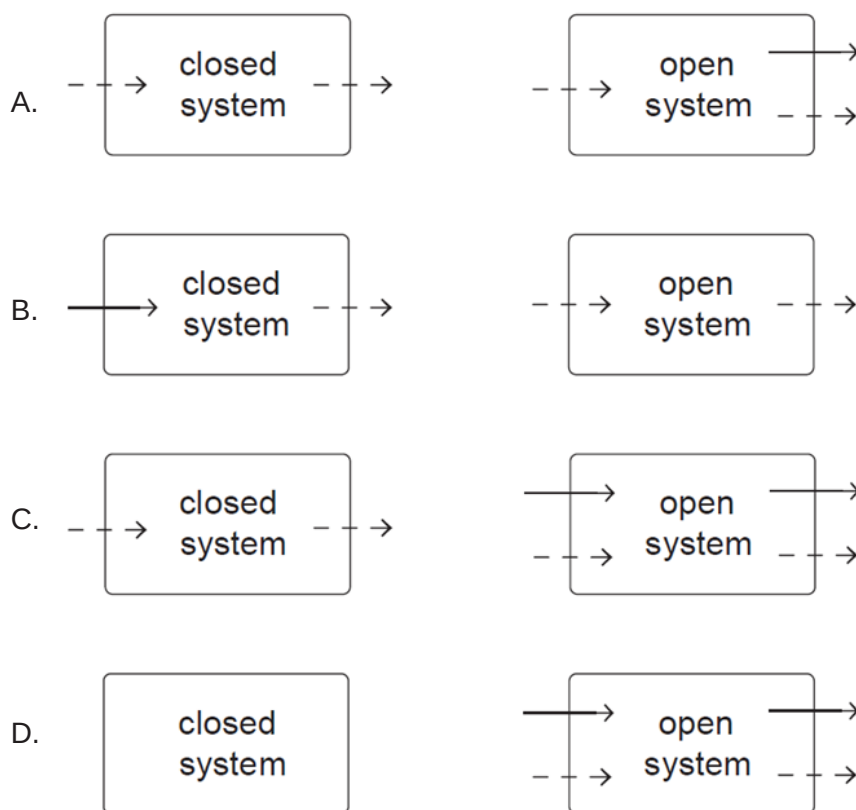


- A. 2-chloro-3-fluorobutane
  - B. 2-fluoro-3-chlorobutane
  - C. 2-dimethyl-1-chloro-2-fluoroethane
  - D. 1,2-dimethyl-1-fluoro-2-chloroethane
8. How many isomers does the formula  $\text{C}_2\text{H}_4\text{Br}_2$  have?
- A. 1
  - B. 2
  - C. 3
  - D. 4
9. 100 mL of 0.4 M nitric acid,  $\text{HNO}_3$ , is added to 100 mL of 0.1 M barium hydroxide,  $\text{Ba}(\text{OH})_2$ . What is the pH of the resulting solution?
- A. 0.3
  - B. 0.7
  - C. 0.8
  - D. 1.0
10. What mass of silver chloride ( $MM = 143.4$ ) will dissolve in 1.00 L of water? The  $K_{sp}$  of  $\text{AgCl}$  is  $1.8 \times 10^{-10}$ .
- A. 1.4 mg
  - B. 1.9 mg
  - C. 2.9 mg
  - D. 3.8 mg

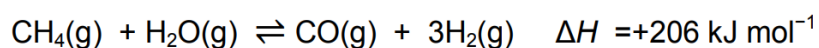
11. A student drew a diagram that illustrated the flow of matter and energy in closed and open systems. They used the following key.



Which of the following could be the student's diagram?



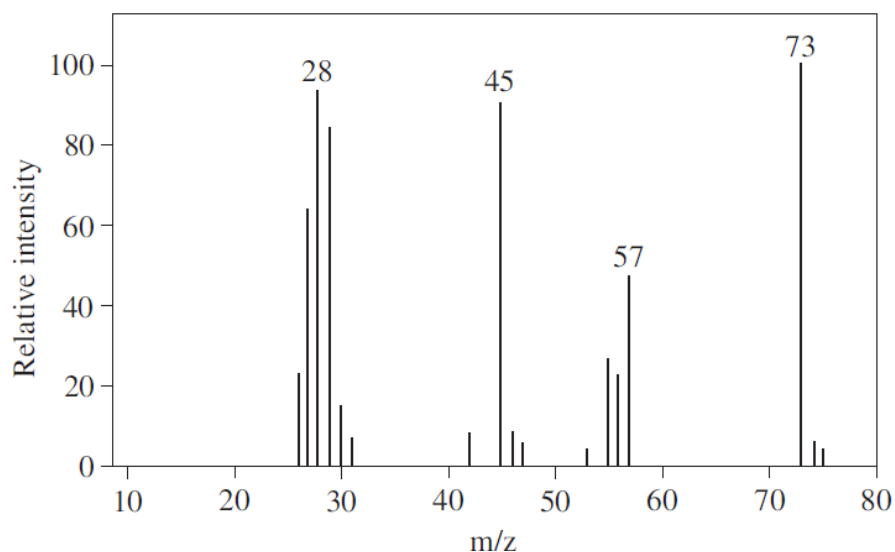
12. The production of hydrogen occurs by the reaction of methane with steam. The reaction mixture reaches a state of dynamic equilibrium.



Which of the following shows how the equilibrium yield of hydrogen and the value of the equilibrium constant are affected by the changes shown?

|    | Change                                 | Effect on equilibrium yield of $\text{H}_2(\text{g})$ | Effect on value of $K_{\text{eq}}$ |
|----|----------------------------------------|-------------------------------------------------------|------------------------------------|
| A. | Increase pressure                      | decrease                                              | decrease                           |
| B. | Add a catalyst                         | increase                                              | no effect                          |
| C. | Increase temperature                   | increase                                              | increase                           |
| D. | Remove $\text{CO}(\text{g})$ as formed | increase                                              | increase                           |

13. A simplified mass spectrum of butanoic acid is shown.

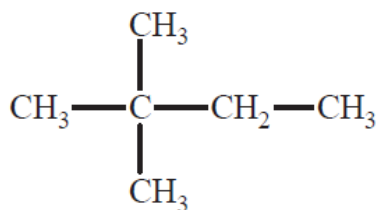


What does the peak with the mass-to-charge ratio,  $m/z$ , of 73 represent?

- A. The greatest mass-to-charge ratio.
- B. The species  $\text{CH}_2\text{CH}_2\text{COOH}^-$
- C. The species  $\text{CH}_2\text{CH}_2\text{COOH}$
- D. The species  $\text{CH}_2\text{CH}_2\text{COOH}^+$

14. The structure of 2,2-dimethylbutane ( $\text{C}_6\text{H}_{14}$ ) is shown below.

In a particular chlorination reaction, a single hydrogen atom of 2,2-dimethylbutane is replaced by one chlorine atom. More than one compound with the formula  $\text{C}_6\text{H}_{13}\text{Cl}$  can be formed.

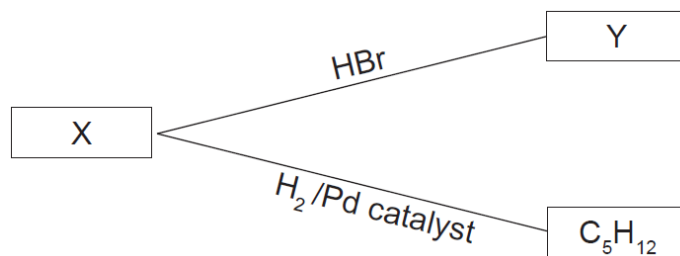


How many different compounds could be formed in this monosubstitution reaction?

- A. 2
- B. 3
- C. 4
- D. 5

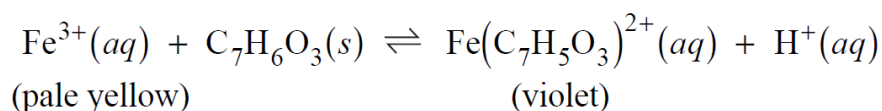
15. The diagram shows two reactions.

Determine the molecular formulae of substances **X** and **Y**.



|    | <b>X</b>                       | <b>Y</b>                                       |
|----|--------------------------------|------------------------------------------------|
| A. | C <sub>5</sub> H <sub>12</sub> | C <sub>5</sub> H <sub>13</sub> Br              |
| B. | C <sub>5</sub> H <sub>10</sub> | C <sub>5</sub> H <sub>11</sub> Br              |
| C. | C <sub>5</sub> H <sub>10</sub> | C <sub>5</sub> H <sub>10</sub> Br <sub>2</sub> |
| D. | C <sub>5</sub> H <sub>8</sub>  | C <sub>5</sub> H <sub>9</sub> Br               |

16. Iron (III) forms a variety of coloured complex ions. One such example is that produced by the reaction with salicylic acid (C<sub>7</sub>H<sub>6</sub>O<sub>3</sub>):



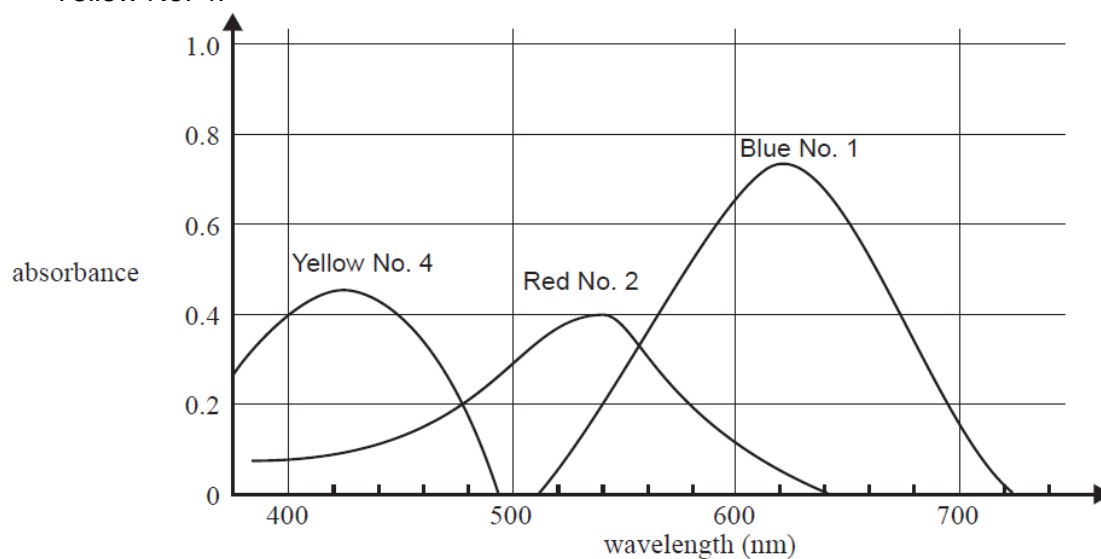
Which of the following will decrease the yield of the violet complex, Fe(C<sub>7</sub>H<sub>5</sub>O<sub>3</sub>)<sup>2+</sup>?

- A. Lowering the pH of the solution
- B. Raising the pH of the solution
- C. Adding C<sub>7</sub>H<sub>6</sub>O<sub>3</sub>(s) to the solution
- D. Adding Fe<sup>3+</sup> ions to the solution

17. Which of the following combinations of solutions will form the most effective buffer solution?

- A. 20 mL of 0.3 mol L<sup>-1</sup> CH<sub>3</sub>COOH and 10 mL of 0.2 mol L<sup>-1</sup> NaCH<sub>3</sub>COO
- B. 10 mL of 0.3 mol L<sup>-1</sup> H<sub>2</sub>O and 20 mL of 0.2 mol L<sup>-1</sup> H<sub>3</sub>O<sup>+</sup>
- C. 20 mL of 0.3 mol L<sup>-1</sup> NaOH and 10 mL of 0.3 mol L<sup>-1</sup> NaCH<sub>3</sub>COO
- D. 20 mL of 0.3 mol L<sup>-1</sup> H<sub>2</sub>CO<sub>3</sub> and 20 mL of 0.3 mol L<sup>-1</sup> NaHCO<sub>3</sub>

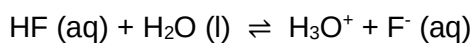
18. The graph shows the absorption spectra of three food dyes: Blue No. 1, Red No. 2 and Yellow No. 4.



Which one of the following is the best wavelength to determine the concentration of Red No.2 dye in a solution containing a mixture of all three dyes?

- A. 430 nm
- B. 500 nm
- C. 540 nm
- D. 620 nm

19. The equation shows the reaction of HF with water.

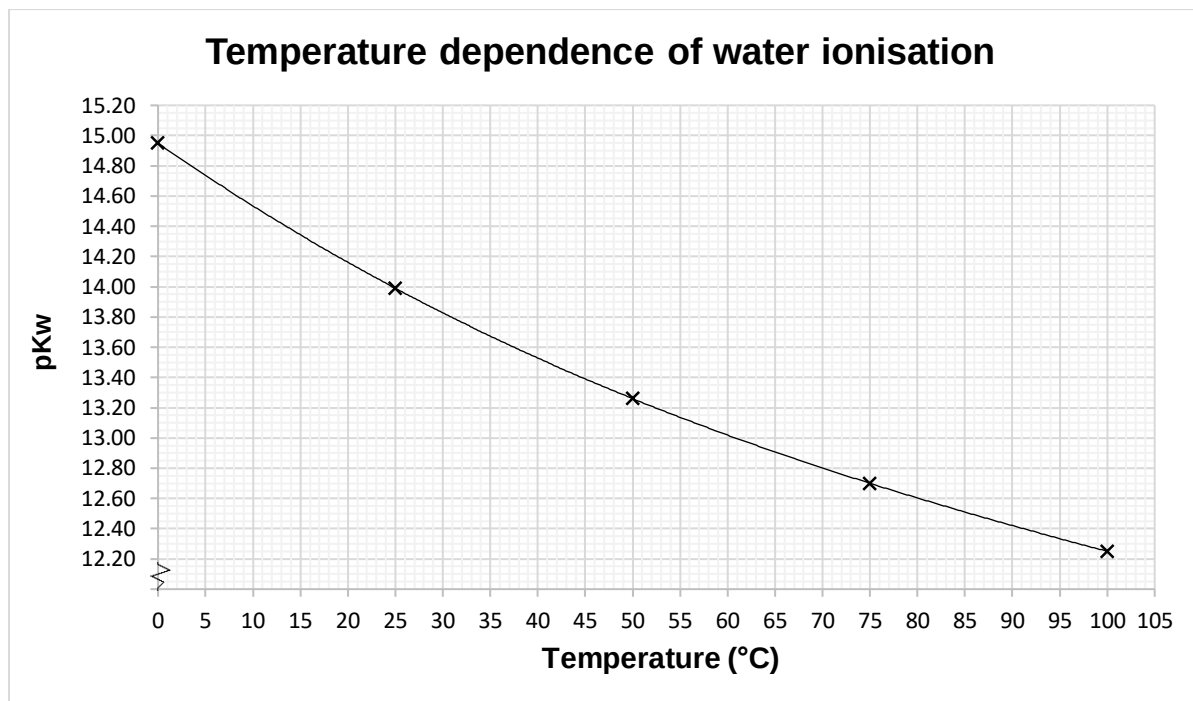


A 0.13M solution of HF has a pH of 2.05.

What is the  $\text{pK}_\text{b}$  of  $\text{F}^-$  ?

- A. 11.81
- B. 2.19
- C. 3.21
- D. 10.79

20. Temperature can affect the value of the equilibrium constant for the ionisation of water ( $K_w$ ). The graph below shows values for  $pK_w$  for temperatures between  $0^\circ\text{C}$  and  $100^\circ\text{C}$ .



A student placed solutions in a water bath at  $60^\circ\text{C}$  to determine if temperature affected the resulting pH of the following reaction.

10.0mL of  $0.15\text{ mol L}^{-1}$  KOH was added to 5mL of distilled water. Predict the final pH of the solution at  $60^\circ\text{C}$ .

- A. 10.50
- B. 12.00
- C. 13.00
- D. 13.17

# Chemistry

## Section II Answer Booklet

**80 marks**

**Attempt Questions 21 – 32**

**Allow about 2 hours and 25 minutes for this section**

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**Instructions**

- Answer the questions in the spaces provided. These spaces provide guidance for the expected length of response.
  - Show all relevant working in questions involving calculations.
  - Extra writing space is provided at the back of this booklet. If you use this space, clearly indicate which question you are answering.
- 

**Please turn over**

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**Question 21** (5 marks)

Molecules of propan-2-ol and propanone each contain three carbon atoms.

- (a) Complete **Table 1** to suggest the shape around the central C atom in a molecule of each compound. **2**

| <b>Table 1</b>                     |                                                             |                                         |
|------------------------------------|-------------------------------------------------------------|-----------------------------------------|
| <b>Compound</b>                    | Propan-2-ol<br>$\text{CH}_3\text{CH}(\text{OH})\text{CH}_3$ | Propanone<br>$\text{CH}_3\text{COCH}_3$ |
| <b>Shape around central C atom</b> |                                                             |                                         |

- (b) Explain why propanone has a lower boiling point than propan-2-ol. **3**

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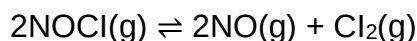
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**Question 22** (4 marks)

Consider the equilibrium reaction:



- (a) Predict which reaction (forward or reverse) favours maximum entropy. Explain your answer. **2**

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- (b) Predict whether the forward reaction is exothermic or endothermic. Explain your answer. **2**

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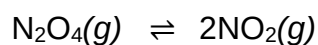
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**Question 23** (6 marks)

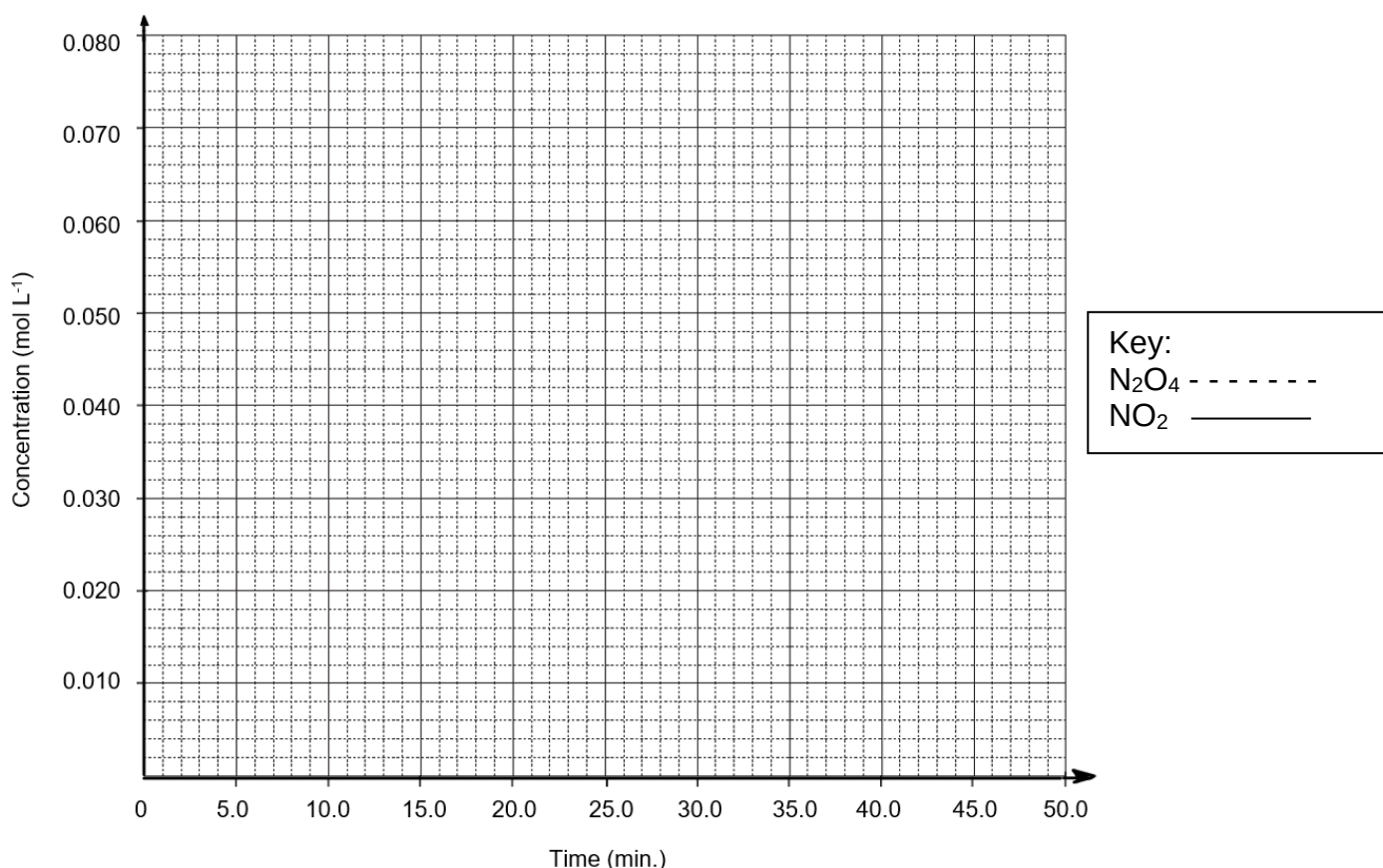
Consider the following equilibrium reaction:



A 2.0 L container is filled with 0.070 mol of  $\text{N}_2\text{O}_4$ . Equilibrium is reached after 20.0 minutes at which time there is 0.040 mol  $\text{N}_2\text{O}_4$  and 0.060 mol of  $\text{NO}_2$ .

- (a) Sketch and label a graph below for the changes in concentrations of  $\text{N}_2\text{O}_4$  and  $\text{NO}_2$  for the time period 0 to 30.0 minutes.

**3**

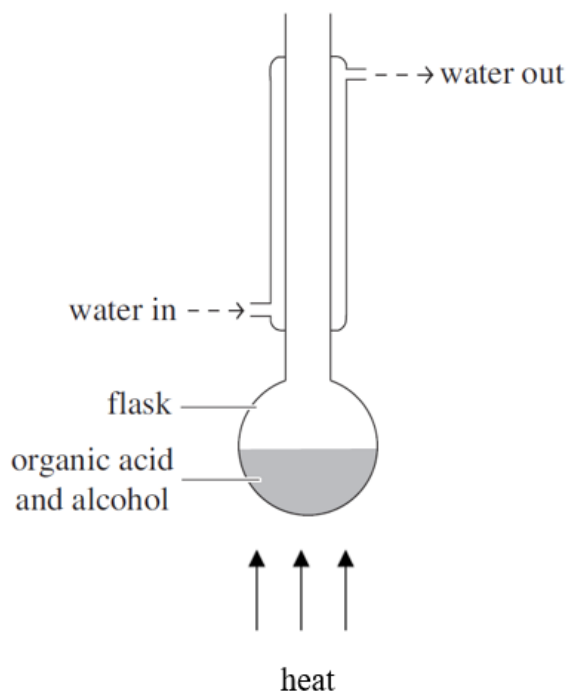


- (b) At time 30.0 minutes a malfunction with the equipment causes the volume to double and the equilibrium is disturbed. Equilibrium is reached after 10 minutes. Sketch and label a graph above for the changes in concentrations of  $\text{N}_2\text{O}_4$  and  $\text{NO}_2$  for the time period 30.0 to 50.0 minutes.

**3**

**Question 24** (8 marks)

Propan-1-ol can react with methanoic acid to produce an organic molecule. The diagram below shows the apparatus used for this reaction.



(a) Identify the type of reaction. 1

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(b) Write a balanced structural chemical equation for the reaction. 2

(c) Identify an appropriate source of heat. Justify your answer. 2

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**Question 24 continues on page 16**

**Question 24** (continued)

- (d) You have performed this experiment in your study of Chemistry. Explain two steps you used to purify your product in the science laboratory. Refer to reagents and equipment used. **3**

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**Question 25** (8 marks)

A student was given 4 colourless liquids that were labelled **A**, **B**, **C** and **D**. They were known to be ethanol, ethanoic acid, pentane and hex-3-ene, but the exact identity of each liquid was unknown.

The student tested the properties of each liquid and obtained the following results.

|                                                                      | <b>A</b>          | <b>B</b>              | <b>C</b>              | <b>D</b>              |
|----------------------------------------------------------------------|-------------------|-----------------------|-----------------------|-----------------------|
| Solubility in water                                                  | insoluble         | soluble               | soluble               | insoluble             |
| Addition of red-coloured bromine solution (aq) (absence of UV light) | colour disappears | no immediate reaction | no immediate reaction | no immediate reaction |
| Addition of sodium carbonate solution (aq)                           | no reaction       | gas evolved           | no reaction           | no reaction           |

(a) Identify each of the liquids.

**2**

**A** = .....

**C** = .....

**B** = .....

**D** = .....

(b) Identify the type of reaction occurring between compound **A** and bromine solution **and** write a balanced structural equation. **3**

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(c) Discuss the different solubilities of the 4 compounds in water.

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**Question 26** (8 marks)

A student was investigating the properties of acids and constructed the following table.

| Acid                       | Concentration<br>(mol L <sup>-1</sup> ) | Type of<br>acid | Available<br>protons | pH at 25°C |
|----------------------------|-----------------------------------------|-----------------|----------------------|------------|
| Hydrocyanic acid<br>(HCN)  | 0.50                                    | weak            | monoprotic           | 4.72       |
| Hydrochloric acid<br>(HCl) | 0.50                                    |                 | monoprotic           |            |

(a)

(i) Complete the table.

**1**

(ii) Calculate the  $K_a$  of Hydrocyanic acid.

**3**

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(b) A buffer solution is prepared by combining 100 mL of 0.010 M hypochlorous acid (HClO) and 0.107 g of solid NaClO.  $K_a \text{ HClO} = 3.0 \times 10^{-8}$ .

Calculate the pH of this buffer solution and explain the classification of the solution as a buffer using appropriate equation/s in your answer.

**4**

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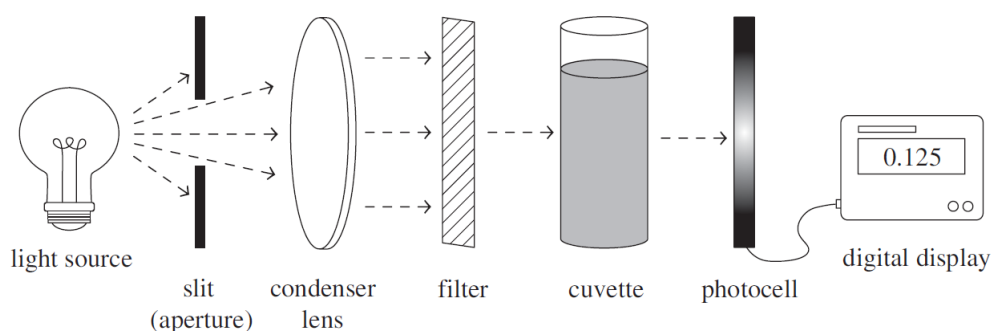
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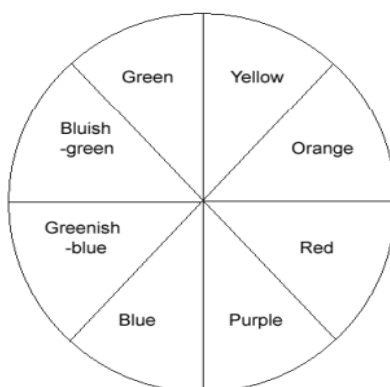
### Question 27 (11 marks)

The diagram shows the major components of a colourimeter.



The concentration of phosphate in a water sample can be determined using a colorimeter. Phosphate is colourless, it can be reacted with ammonium molybdate and ascorbic acid in the presence of sulfuric acid to form a blue complex.

An analytical chemist used a colour wheel to select a suitable filter to determine the phosphate concentration of an industrial effluent water sample.



- (a) Identify the most appropriate colour for the filter in the colorimeter for this analysis.

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Question 27 continues on page 20

**Question 27 (continued)**

Solutions of known phosphate concentration were used to calibrate the colourimeter, so that absorbance was known for the concentrations shown in the table below.

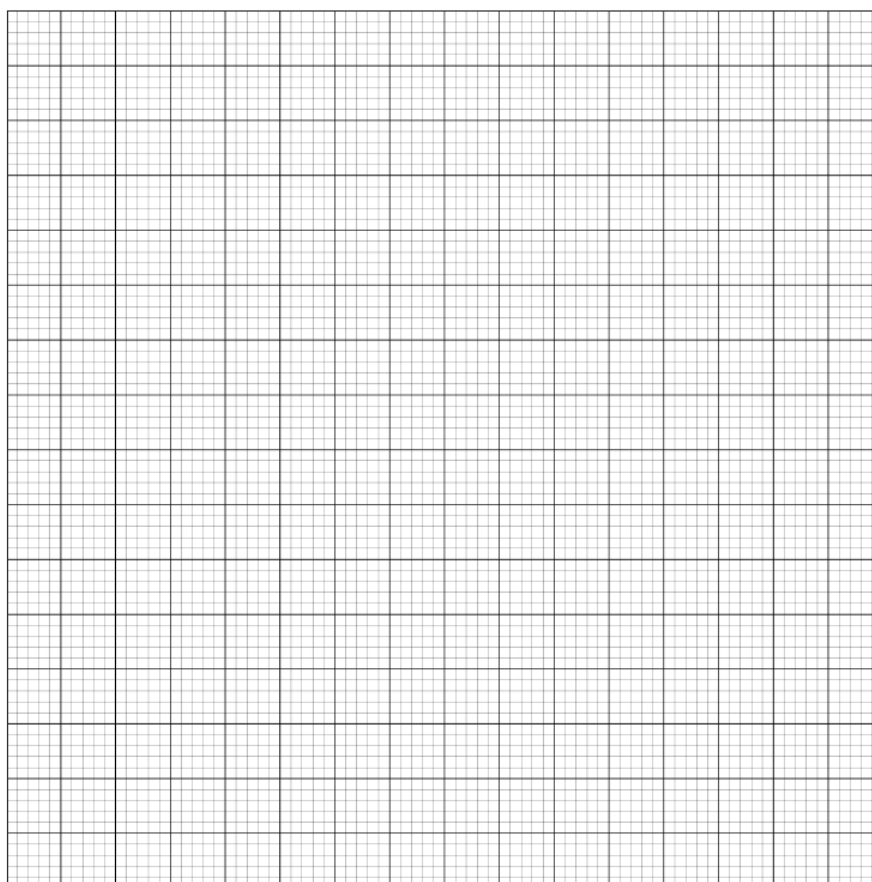
- (b) Graph the data given in Table 1 using the grid below. Draw a straight line of best fit.

**3**

**Table 1**  
**Absorbance values for phosphate solutions of known concentration**

| Concentration<br>mg/L | Absorbance |
|-----------------------|------------|
| 0.00                  | 0.003      |
| 0.15                  | 0.095      |
| 0.30                  | 0.194      |
| 0.45                  | 0.290      |
| 0.60                  | 0.378      |
| 0.90                  | 0.573      |
| 1.20                  | 0.745      |

Calibration Curve of Phosphate Concentrations vs Absorbance



**Question 27 continues on page 21**

**Question 27 (continued)**

Environment Protection Authority (EPA) recommends that effluent from industrial sites entering a waterway should have a limit of  $5.0 \times 10^{-7} \text{ mol L}^{-1}$  of phosphate to avoid algae growth.

10 mL of a filtered effluent sample was placed in a 100 mL volumetric flask and made up to the mark with distilled water. This sample was tested using the colourimeter. The absorbance value for this solution was 0.4440.

- (c) Use the graphed data to determine the concentration of phosphate in the effluent and determine if the effluent water complies with EPA recommendations. **3**

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- (d) As a precaution, phosphate can be reduced in effluent wastewater before it enters waterways using precipitation.

In the research laboratory 100 mL of the wastewater containing  $2.1 \times 10^{-4} \text{ mol L}^{-1}$  of phosphate ions is tested.

Will a precipitate of calcium phosphate form when the wastewater sample is added to 200 mL of  $5 \times 10^{-2} \text{ mol L}^{-1}$  of calcium chloride?

Justify your answer. **4**

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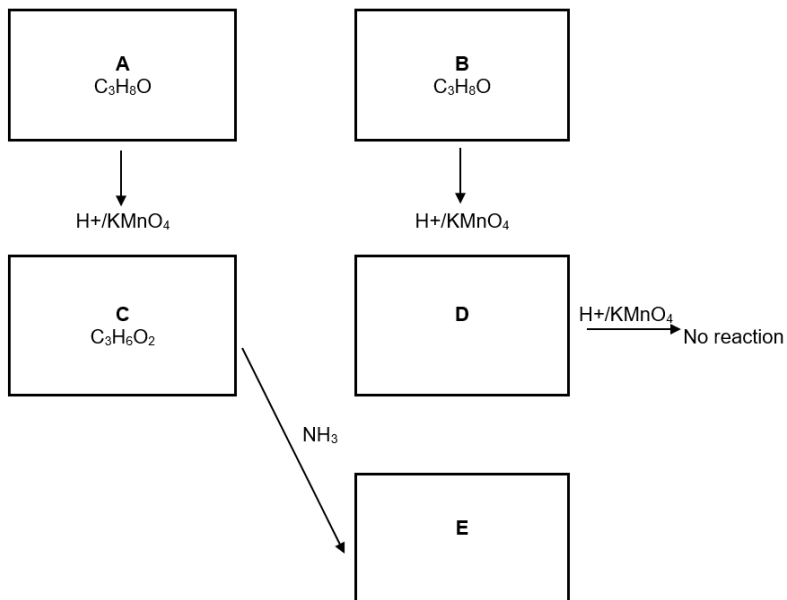
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**Question 28** (5 marks)

Two different compounds A and B are isomers with the molecular formula  $C_3H_8O$ .

A and B undergo a series of reactions as shown below.

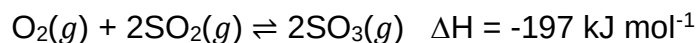


Draw the structure of each compound, A to E, in the corresponding space provided.

|   |   |
|---|---|
| A | B |
| C | D |
|   | E |

**Question 29** (6 marks)

Consider the following equilibrium reaction:



- (a) 2.0 moles of oxygen was heated with 0.89 moles of sulfur dioxide in a 2.0 L sealed container at 25°C. When equilibrium was established 0.80 moles of sulfur trioxide was present.

Calculate  $K_{\text{eq}}$  at 25°C.

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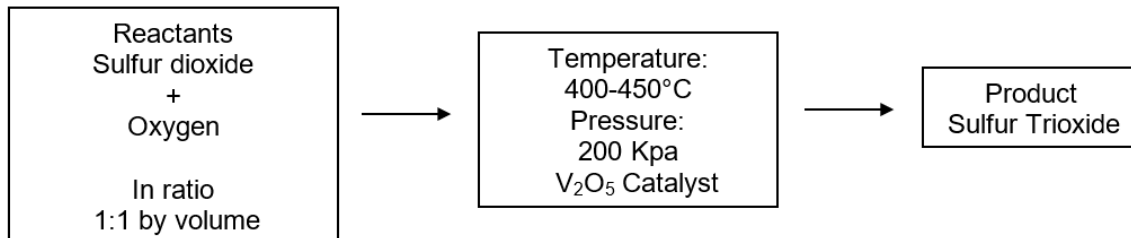
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In the industrial process used to produce sulfur trioxide the conditions shown below are used.



- (b) Analyse the conditions chosen by Engineers for this chemical synthesis process.

**3**

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### Question 30 (8 marks)

The formulae of two isomers, labelled compounds J and K, are shown below:

| Compound J                                                                                                                                                                                                                                                                                                                                                        | Compound K                                                                                                                                                                                                                                                                                                                                |
|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| $  \begin{array}{ccccccc}  & & & \text{O} & & & \\  & & &    & & & \\  \text{H} & - & \text{C} & - & \text{O} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{H} \\    & & & & & &   & &   & & & & \\  \text{H} & & & & & & \text{H} & & \text{H} & & & & \\  & & & & & &   & &   & & & & \\  & & & & & & \text{H} & & \text{H} & & & &   \end{array}  $ | $  \begin{array}{ccccccc}  & & & \text{H} & & & \\  & & &   & & & \\  & & \text{H} & - & \text{C} & - & \text{H} & & \text{O} & \\  & & &   & & & & &    & \\  \text{H} & - & \text{C} & - & \text{C} & - & \text{C} & - & \text{O} & - & \text{H} \\    & & &   & & & & & & & \\  \text{H} & & & \text{H} & & & & & & &   \end{array}  $ |

- (a) Identify the type of isomer shown above. **1**

A student was provided the mass spec, IR and  $^{13}\text{C}$  NMR for both compounds.

- (b) Identify the spectra that belongs to compound J and K in the title section on the following page (page 25). 1

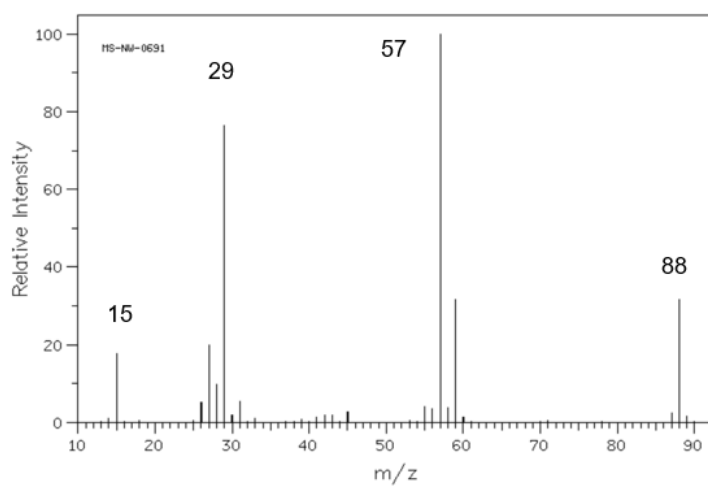
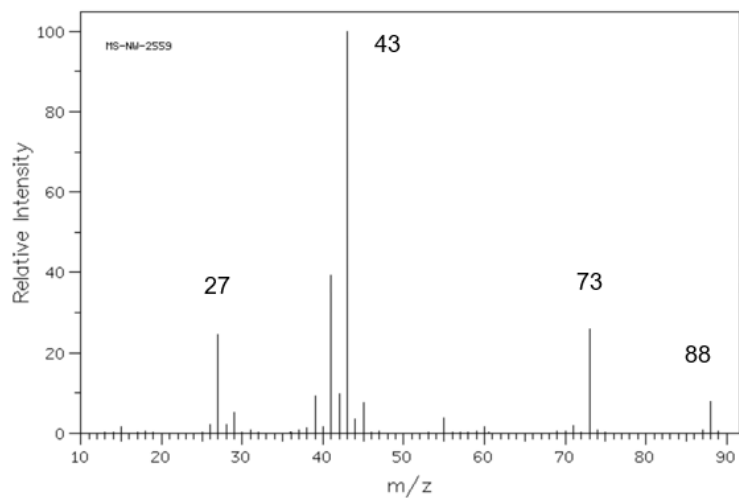
- (c) Explain your reasoning referencing all spectra data on the following page. **6**

This image shows a full page of white paper with horizontal dotted lines, resembling notebook paper. The lines are evenly spaced and run across the width of the page. There are no margins, text, or other markings present.

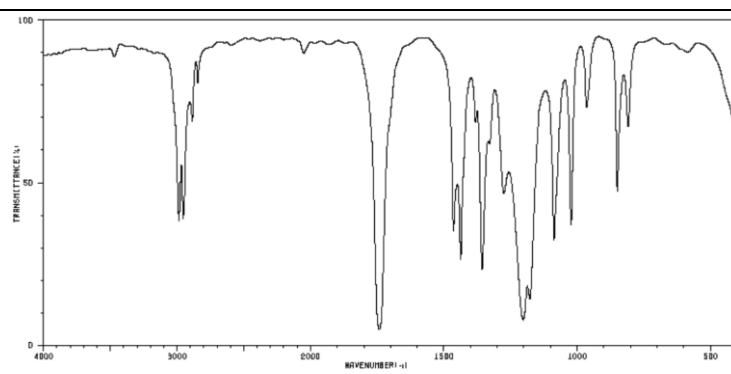
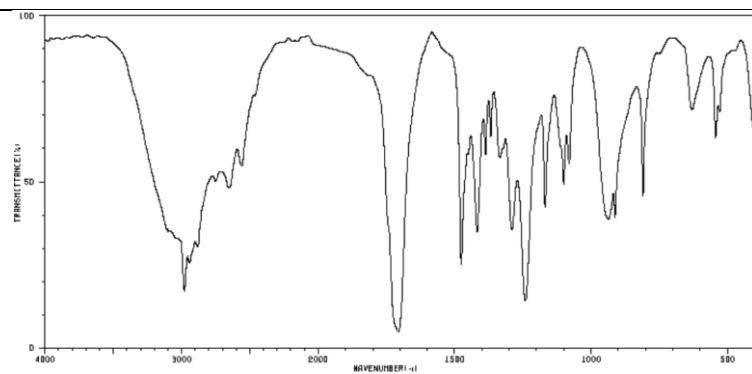
Compound \_\_\_\_\_

Compound \_\_\_\_\_

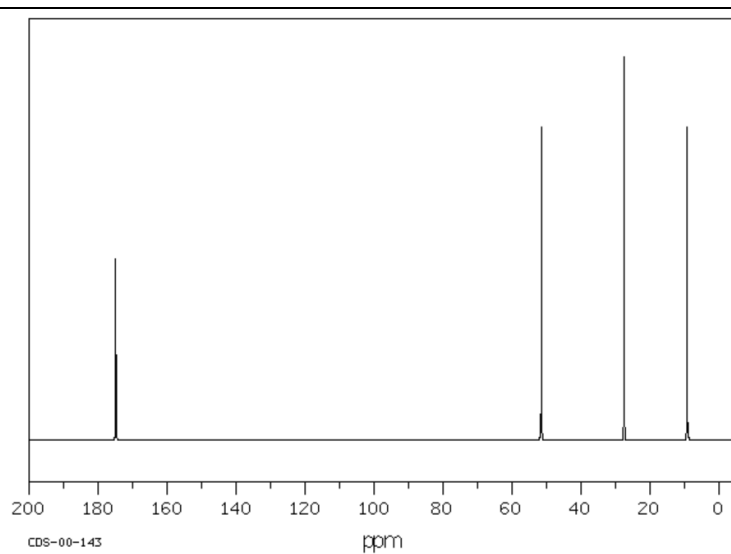
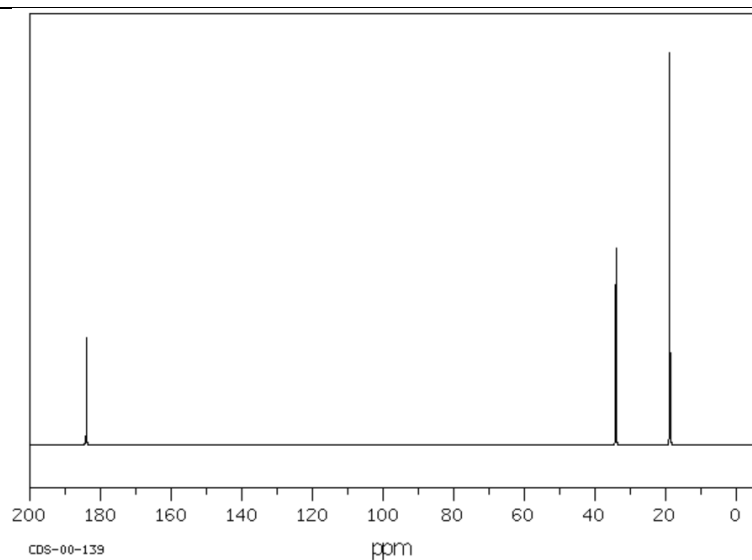
### Mass Spectrometry



### Infra Red Spectroscopy

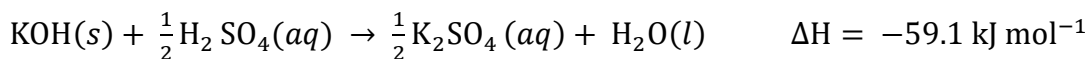


### C13 NMR



**Question 31** (5 marks)

Some thermochemical data relating to potassium hydroxide is shown below.



An 80.0 mL sample of 0.50 mol L<sup>-1</sup> H<sub>2</sub>SO<sub>4</sub> was poured into a poly styrofoam cup. A temperature probe placed into the acid recorded an initial temperature of 24.3°C.

5.00 g of KOH was then added to the acid and completely dissolved in it. The probe was used to determine the highest temperature reached upon this dissolution.

- (a) Show by calculation which is the limiting reagent in the reaction between potassium hydroxide and sulfuric acid. **2**

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- (b) Calculate the highest theoretical temperature which could be measured by the probe. Assuming no energy loss to the environment. **3**

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**Question 32** (6 marks)

Evaluate the following statement. Refer to provided data and use appropriate chemical equations.

*“Biofuels are a better option than traditional fossil fuels as an energy source.”*

| Fuel            | Formula                             | $\Delta H^\circ_c$ kJ/g |
|-----------------|-------------------------------------|-------------------------|
| Octane (petrol) | $C_8H_{18}$                         | 48                      |
| Diesel          | $C_{12}H_{26}$                      | 44                      |
| Biodiesel       | $C_{11}H_{26}O_2 - C_{19}H_{34}O_2$ | 40 – 48*                |
| Ethanol         | $C_2H_5OH$                          | 29.78                   |

\* NOTE: Biodiesel can have varying chain lengths.

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**End of paper**

## Section II extra writing space

**If you use this space, clearly indicate which questions you are answering.**

[illegible]

## Section II extra writing space

**If you use this space, clearly indicate which questions you are answering.**

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Student Number: .....

### Chemistry Multiple Choice Answer Sheet

Select the alternative A, B, C or D that best answers the question.

Fill in the response space completely. If you think you have made a mistake, put a cross through the incorrect answer and fill in the new answer.

If you change your mind and have crossed out what you consider to be the correct answer, then indicate this by writing the word **correct** and drawing an arrow.

| Start Here | 1.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|------------|-----|-------------------------|-------------------------|-------------------------|-------------------------|
|            | 2.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 3.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 4.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 5.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 6.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 7.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 8.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 9.  | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 10. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 11. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 12. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 13. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 14. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 15. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 16. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 17. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 18. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 19. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |
|            | 20. | A <input type="radio"/> | B <input type="radio"/> | C <input type="radio"/> | D <input type="radio"/> |

# 2023 CHEMISTRY TRIAL MARKING GUIDELINES

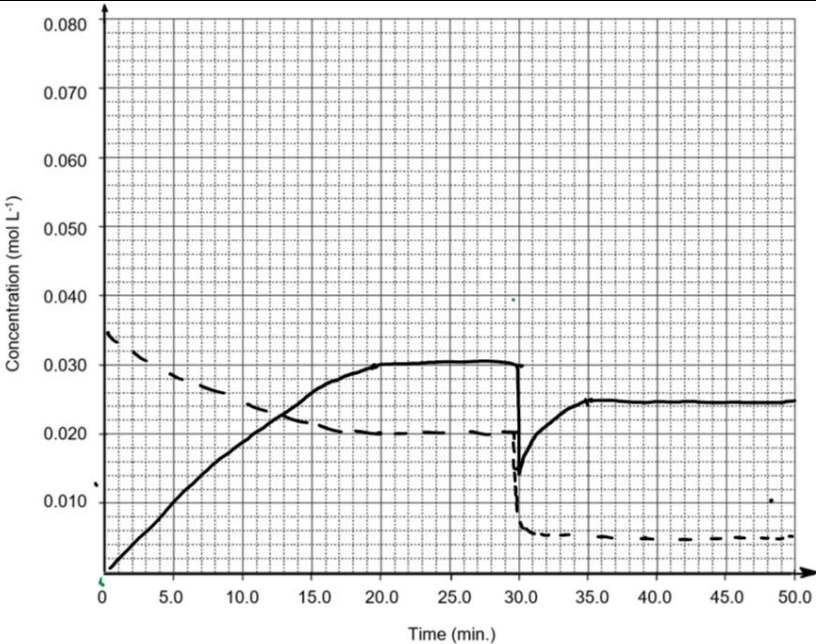
## MULTIPLE CHOICE

|          |   |   |   |   |   |   |   |   |   |    |    |    |    |    |    |    |    |    |    |    |
|----------|---|---|---|---|---|---|---|---|---|----|----|----|----|----|----|----|----|----|----|----|
|          | 1 | 2 | 3 | 4 | 5 | 6 | 7 | 8 | 9 | 10 | 11 | 12 | 13 | 14 | 15 | 16 | 17 | 18 | 19 | 20 |
|          | A | D | C | A | B | C | A | B | D | B  | C  | C  | D  | B  | B  | A  | D  | B  | D  | B  |
| Catch up |   |   |   |   | B |   | D | C |   |    |    | B  |    |    |    |    |    |    |    | A  |

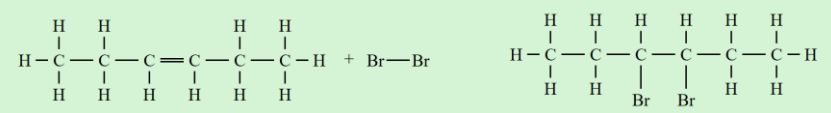
|       | Marking Guideline                                                                                                                            |        | Suggested Answer                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 21 a) | Criteria                                                                                                                                     | Marks  | <p>Propan-2-ol - tetrahedral</p> <p>Propanone – trigonal planar</p> <p>Must be named (not drawn)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Generally answered well. Some errors included drawing the structural formula, no marks were awarded for this. A few students left the question blank.</p> <p>If you identified propanone using trigonal this was not awarded as trigonal can be planar and pyramidal.</p> <p>Also a few students named them back to front, no marks were awarded in this case.</p>                                                                                                                                                                                                      |
|       | <ul style="list-style-type: none"> <li>correct shape around central carbon identified for both molecules</li> </ul>                          | 2      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|       | <ul style="list-style-type: none"> <li>1 correct shape identified</li> </ul>                                                                 | 1      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 21 b) | Criteria                                                                                                                                     | Marks  | <ul style="list-style-type: none"> <li>Identifies most significant IMF for propanone and propan-2-ol</li> <li>Explains difference in force strength.</li> <li>Link to weaker force to less energy required to boil, or stronger force requires more energy to boil.</li> </ul> <p>The boiling points are determined by the intermolecular forces between their molecules. Both molecules have dispersion forces. However, propanone can form dipole-dipole forces due to the carbonyl group and propan-2-ol can form H-bonds due to the presence of hydroxyl group. As dipole-dipole is a weaker force compared to hydrogen bonding this results in propanone having a lower boiling point as therefore, less energy is required to overcome the attractive force.</p> | <p>Answered well across cohort. Some stated hydrogen bond between O-H in alcohol, note this is a covalent bond. Hydrogen bond is between the molecules.</p> <p>The O-H bond does not break, if you said this you lost marks.</p> <p>Other marks lost is no mention of dipole-dipole.</p> <p>Another error was indicating both molecules have hydrogen bonding, and the difference was due to dispersion forces. Note the MM of each molecule is very similar, and therefore the dispersion forces will be similar. It is important to discuss the most significant IMF</p> |
|       | <ul style="list-style-type: none"> <li>Provides thorough explanation of difference with reference to H-bonding and dipole-dipole.</li> </ul> | 3      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
|       | <ul style="list-style-type: none"> <li>Provides some explanation</li> <li>Some relevant information</li> </ul>                               | 2<br>1 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |

|          |                                                                                                                                                                          |       |                                                                                                                                                   |                                 |
|----------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------|
| 22<br>a) | Criteria                                                                                                                                                                 | Marks | The forward reaction favours maximum entropy. The RHS has more moles of gas, hence there is more disorder on the RHS, and therefore more entropy. | Answered well across the cohort |
|          | <ul style="list-style-type: none"> <li>Identifies correct direction.<br/>AND</li> <li>Links increasing disorder with number of moles on each side of equation</li> </ul> | 2     |                                                                                                                                                   |                                 |
|          | <ul style="list-style-type: none"> <li>Identifies correct direction.</li> </ul>                                                                                          | 1     |                                                                                                                                                   |                                 |

|          |                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                              |
|----------|---------------------------------------------------------------------------------------------------------------------------------------|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22<br>b) | Criteria                                                                                                                              | Marks | Endothermic. Since the <b>forward</b> reaction favours maximum entropy, the <b>reverse</b> reaction must favour minimum enthalpy.<br><br><b>Needs to relate to Gibbs for two marks.</b><br><b>Or linking the concept or Gibbs/ spontaneity</b><br>Using words to explain:<br>e.g for an $\rightleftharpoons$ reaction the driving force for one direction is entropy and the opposite direction would be enthalpy. So that $\Delta G$ can be zero for in both directions $\rightleftharpoons$<br><br>Using equation to explain:<br>$\Delta G = \Delta H - T \Delta S$<br>Therefore, as the driving force is a + entropy for the forward reaction the enthalpy must be positive i.e., endothermic.<br>OR<br>At equilibrium $\Delta G = 0$<br>If $\Delta S$ is positive $\Delta H$ must also be positive for $\Delta G$ to equal 0 | If answered endothermic and then went on to explain more energy absorbed in breaking bonds compared to energy released in making bonds. Did not award the second mark. You need data to calculate overall $\Delta H$ . You were not given this data.<br>The explanation needed to use Gibb's |
|          | <ul style="list-style-type: none"><li>Makes correct prediction.<br/>AND</li><li>Provides a correct reason related to Gibbs.</li></ul> | 2     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                              |
|          | <ul style="list-style-type: none"><li>Makes correct prediction.<br/>OR</li><li>Use Gibbs equation with wrong conclusion.</li></ul>    | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                                                                                                              |
| 23<br>a) |                                                                                                                                       |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | Some students did not account for the volume 2L. Initial moles and moles at equilibrium were provided.<br>Only penalised once if this was the case.<br>Other errors, not starting $\text{NO}_2$ at 0, not showing $\text{N}_2\text{O}_4$ decreasing slope, some used a straight line.        |

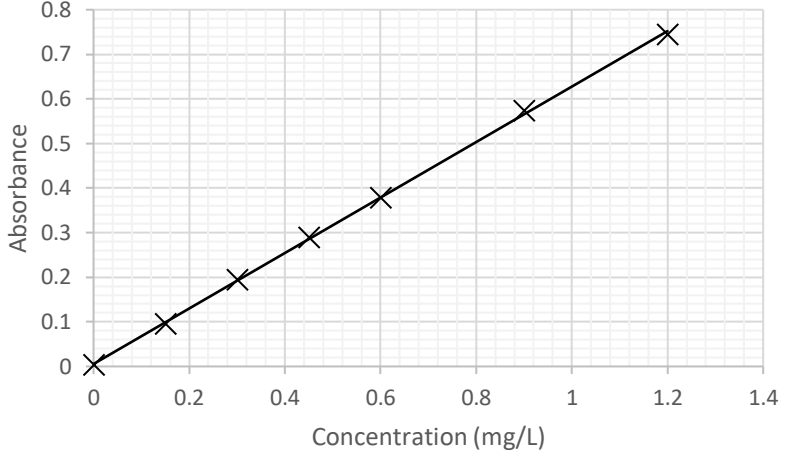
|              |                                                                                                                                                                                                                                                                                                                                                                                                        |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|--------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|              | Criteria                                                                                                                                                                                                                                                                                                                                                                                               | Marks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|              | <ul style="list-style-type: none"> <li>Correct concentrations for both reactants</li> <li>Correct equilibrium concentration for both reactants</li> <li>Equilibrium lines drawn constant concentration and correct time</li> </ul>                                                                                                                                                                     | 3     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|              | <ul style="list-style-type: none"> <li>2 of above correct</li> </ul>                                                                                                                                                                                                                                                                                                                                   | 2     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|              | <ul style="list-style-type: none"> <li>1 of above correct</li> </ul>                                                                                                                                                                                                                                                                                                                                   | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
| <b>23 b)</b> | Criteria                                                                                                                                                                                                                                                                                                                                                                                               | Marks | <p>See diagram above from 30min</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|              | <ul style="list-style-type: none"> <li>Spike for <math>\text{N}_2\text{O}_4</math> and <math>\text{NO}_2</math> down</li> <li>Spike drops by the correct amount.</li> <li>2 correct curves (i.e. <math>[\text{NO}_2]</math> rises after spike, and <math>\text{N}_2\text{O}_4</math> drops).</li> <li>line must not go above original level.</li> <li>1:2 ratio of change in concentrations</li> </ul> | 3     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|              | <ul style="list-style-type: none"> <li>3-4 of the above but must include.</li> <li>Spike for <math>\text{N}_2\text{O}_4</math> and <math>\text{NO}_2</math> down</li> <li>2 correct curves (i.e. <math>[\text{NO}_2]</math> rises after spike, and <math>\text{N}_2\text{O}_4</math> drops).</li> </ul>                                                                                                | 2     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|              | <ul style="list-style-type: none"> <li>Shows some understanding of sketching equilibrium curves</li> </ul>                                                                                                                                                                                                                                                                                             | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|              |                                                                                                                                                                                                                                                                                                                                                                                                        |       | <p>Part b was not answered as well as part a. Common error was not understanding the effect of volume change on concentrations or not showing the correct concentrations after the change. If the volume is doubled the concentration will halve. ECF was applied from part b as long as the concentrations decreased by half at 30 minutes.</p> <p>Some had both <math>\text{NO}_2</math> and <math>\text{N}_2\text{O}_4</math> shift in the same direction after disturbance. Other errors included the incorrect 2:1 ratio, the concentration of <math>\text{NO}_2</math> higher than the original level and equilibrium not being re-established by 40 minutes.</p> |

|          |                                                                                                                                                                                                                                   |                   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 24<br>a) | Criteria                                                                                                                                                                                                                          | Marks             | <ul style="list-style-type: none"> <li>- Esterification</li> <li>- Accept condensation</li> </ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|          | <ul style="list-style-type: none"> <li>Identifies the type of reaction correctly</li> </ul>                                                                                                                                       | 1                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 24<br>b) | Criteria                                                                                                                                                                                                                          | Marks             | <p>Propan-1-ol + methanoic acid <math>\leftrightarrow</math> propyl methanoate + water</p> <div style="background-color: #e0ffe0; padding: 10px; text-align: center;"> <math display="block">  \begin{array}{c}  \text{H} &amp; \text{H} &amp; \text{H} \\    &amp;   &amp;   \\  \text{H}-\text{C}-\text{C}-\text{C}-\text{O}-\text{H} &amp; + &amp; \text{O}=\text{C}-\text{O}-\text{H} \\    &amp;   &amp;   \\  \text{H} &amp; \text{H} &amp; \text{H}  \end{array}  \quad  \begin{array}{c}  \text{H} &amp; \text{H} &amp; \text{H} \\    &amp;   &amp;   \\  \text{H}-\text{C}-\text{O}-\text{C}-\text{C}-\text{C}-\text{H} \\     &amp;   &amp;   &amp;   \\  \text{O} &amp; \text{H} &amp; \text{H} &amp; \text{H}  \end{array}  + \text{H}-\text{O}-\text{H}  </math> </div> <p><math>\text{CH}_3\text{CH}_2\text{CH}_2\text{OH} + \text{C}(\text{O})\text{HCOH} \leftrightarrow \text{C}(\text{O})\text{HCH}_2\text{CH}_2\text{CH}_3 + \text{H}_2\text{O}</math></p> | <p>Better answers included reaction conditions (reflux and conc. <math>\text{H}_2\text{SO}_4</math>)<br/>Students should be using O-H rather than OH to show all the bonds i.e structure<br/><b>Many</b> are losing a mark for no equilibrium arrow. Many are forgetting to include water. (not enough marks to penalise for this and arrows, so 1 mark if forgot both)<br/>Still the problem of “mid air” bonds to OH. The carbon has to be linked directly with the O, not between the O and H. Noted but marks not deducted.</p> |
|          | <ul style="list-style-type: none"> <li>Correct structural equation with equilibrium arrow</li> <li>Incorrect structure OR</li> <li>No equilibrium arrow</li> </ul>                                                                | <p>2</p> <p>1</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 24<br>c) | Criteria                                                                                                                                                                                                                          | Marks             | <p>A hot plate/ mantle is an appropriate source of heat. This prevents overheating – avoiding risks of ignition or explosion because of the flammability of the products and reactants. .</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | <p>Water bath was accepted however hot plate is better choice.<br/>Need to be specific about what the safety risk is, not just a safety risk. Many girls mention the flammability of alcohol. Esters are also highly flammable and volatile.</p>                                                                                                                                                                                                                                                                                    |
|          | <ul style="list-style-type: none"> <li>Identifies an appropriate source of heat</li> <li>Justifies the answer – must relate to safety- flammable substances</li> <li>Identifies an appropriate source</li> </ul>                  | <p>2</p> <p>1</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 24<br>d) | Criteria                                                                                                                                                                                                                          | Marks             | <ol style="list-style-type: none"> <li>Using a separating funnel - Add water to separate the remaining soluble substances from the insoluble ester</li> <li>Add sodium carbonate solution to the ester in a separating funnel to remove any remaining acid (sulfuric and organic)</li> </ol>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <p>Explanation of what the purpose of the steps was missing in many responses. Many forgot to mention equipment (-1)<br/>Several girls just described refluxing, not purifying</p>                                                                                                                                                                                                                                                                                                                                                  |
|          | <ul style="list-style-type: none"> <li>Identifies two steps</li> <li>Explains (including reason for the step) at least 1 step</li> <li>Lists reagents/equipment</li> <li>Explains two steps without specific equipment</li> </ul> | <p>3</p> <p>2</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|       |                                                                                                                                                                                                  |   |                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                         |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|       | <ul style="list-style-type: none"> <li>Or explains one steps with equipment</li> </ul>                                                                                                           |   | 3. Distillation of mixture to separate lower boiling point ester from any further traces of water                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                         |
|       | <ul style="list-style-type: none"> <li>Explains one step OR lists equipment or names 2 processes.</li> </ul>                                                                                     | 1 | 4. Add calcium chloride (or another named dehydrating agent ) to remove any traces of water                                                                                                                                                                                                                                                                                  |                                                                                                                                                                                                         |
| 25 a) | Criteria:<br><ul style="list-style-type: none"> <li>Identifies all four correctly</li> </ul>                                                                                                     | 2 | A- Hex-3-ene<br>B- Ethanoic acid<br>C- Ethanol<br>D- Pentane                                                                                                                                                                                                                                                                                                                 | Very well answered                                                                                                                                                                                      |
|       | <ul style="list-style-type: none"> <li>Identifies 2-3 correctly</li> </ul>                                                                                                                       | 1 |                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                         |
| 25 b) | Criteria:<br><ul style="list-style-type: none"> <li>Identify the type of reaction</li> <li>Writes/ draws a correct structural equation</li> </ul>                                                | 3 | Type of reaction – addition<br>                                                                                                                                                                                                                                                            | Halogenation /bromination not accepted as they are not unique to alkenes and can react with alkanes too. This reaction was able to identify the alkene due to its ability to ADD/ break the double bond |
|       | <ul style="list-style-type: none"> <li>Identify the type of reaction</li> <li>Write a mostly correct structural equation</li> <li>OR correct equation with incorrect type of reaction</li> </ul> | 2 |                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                         |
|       | <ul style="list-style-type: none"> <li>Identify the type of reaction</li> <li>OR</li> <li>Write a mostly correct structural equation</li> </ul>                                                  | 1 |                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                         |
| 25 c) | Criteria:<br><ul style="list-style-type: none"> <li>Discusses the different water solubilities of all FOUR compounds.</li> <li>Identifies the Correct IMF for each compound</li> </ul>           | 3 | Alkanes and alkenes (A and D) are hydrocarbons. They are not generally soluble in water due to being non-polar<br>Student then discusses either i) or ii)<br>i) Weak dispersion forces are present only<br>ii) no hydrogen bond interaction with the water.<br><br>Ethanoic acid (B) is polar and soluble as it can form Hydrogen bonds between the acid molecule and water. | Well answered.<br><br>Must indicate that hydrogen bonds form <b>with</b> water                                                                                                                          |

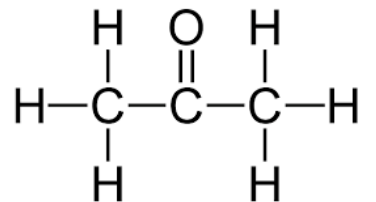
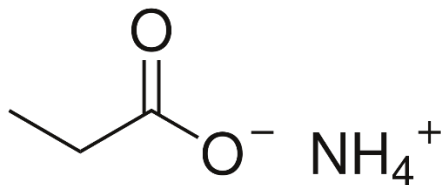
|                                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>Correctly discusses the different water solubilities of at least TWO compounds.</li><li>OR</li><li>Discusses polarity in detail without IMF</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                   | 2                     | Ethanol (C) is polar and so are soluble in water. Hydrogen bonds form between water and ethanol |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------|-------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-------------------------------------------------------------------------------------------------|--------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|------------------------|------|------|------------|------|-------------------------|------|--------|------------|------|-------------------------------------------------------------------------------------------------|---|------------------------|-----------------------|-----------------------|---|----------------------------|----------------------|----------------------|--------------------------------------------------------------------|
|                                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>Correctly discusses the different water solubilities of at least ONE compound.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 1                     |                                                                                                 |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| 26 a) i                                                                                                                                                                                                                           | <table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td><ul style="list-style-type: none"><li>Correct calculation and identification of acid</li></ul></td><td>1</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                            | Criteria:             | Marks                                                                                           | <ul style="list-style-type: none"><li>Correct calculation and identification of acid</li></ul>                                                                                                                                    | 1 | <p>pH = <math>-\log_{10}(0.50)</math><br/>Concentration is 2sigfig so pH should be 2dp</p> <table><tr><th>Acid</th><th>Concentration (mol L<sup>-1</sup>)</th><th>Type of acid</th><th>Available protons</th><th>pH at 25°C</th></tr><tr><td>Hydrocyanic acid (HCN)</td><td>0.50</td><td>weak</td><td>monoprotic</td><td>4.72</td></tr><tr><td>Hydrochloric acid (HCl)</td><td>0.50</td><td>STRONG</td><td>monoprotic</td><td>0.30</td></tr></table> | Acid | Concentration (mol L <sup>-1</sup> )                                                            | Type of acid | Available protons                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | pH at 25°C | Hydrocyanic acid (HCN) | 0.50 | weak | monoprotic | 4.72 | Hydrochloric acid (HCl) | 0.50 | STRONG | monoprotic | 0.30 | Students should be using the pH Sig fig rule which would be 2 dp as concentration in 2 sigfigs. |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| Criteria:                                                                                                                                                                                                                         | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |                                                                                                 |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| <ul style="list-style-type: none"><li>Correct calculation and identification of acid</li></ul>                                                                                                                                    | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |                                                                                                 |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| Acid                                                                                                                                                                                                                              | Concentration (mol L <sup>-1</sup> )                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Type of acid          | Available protons                                                                               | pH at 25°C                                                                                                                                                                                                                        |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| Hydrocyanic acid (HCN)                                                                                                                                                                                                            | 0.50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | weak                  | monoprotic                                                                                      | 4.72                                                                                                                                                                                                                              |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| Hydrochloric acid (HCl)                                                                                                                                                                                                           | 0.50                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | STRONG                | monoprotic                                                                                      | 0.30                                                                                                                                                                                                                              |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| 26 a) ii                                                                                                                                                                                                                          | <table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td><ul style="list-style-type: none"><li>Writes the Ka expression</li><li>Substitutes correct numbers in (does not need an ice table)</li><li>Must make correct statement about negligible change in concentration for HCN</li></ul></td><td>3</td></tr><tr><td><ul style="list-style-type: none"><li>Provides main steps of the calculation does not show all working.</li></ul></td><td>2</td></tr><tr><td><ul style="list-style-type: none"><li>Provides some relevant steps of the calculation</li></ul></td><td>1</td></tr></table> | Criteria:             | Marks                                                                                           | <ul style="list-style-type: none"><li>Writes the Ka expression</li><li>Substitutes correct numbers in (does not need an ice table)</li><li>Must make correct statement about negligible change in concentration for HCN</li></ul> | 3 | <ul style="list-style-type: none"><li>Provides main steps of the calculation does not show all working.</li></ul>                                                                                                                                                                                                                                                                                                                                    | 2    | <ul style="list-style-type: none"><li>Provides some relevant steps of the calculation</li></ul> | 1            | <p>Writes Ka expression<br/>HCN] &lt;-&gt; H+ + CN-<br/>Ka = [H+] [CN-] / ]HCN]</p> <table><tr><td></td><td>HCN]</td><td>H+</td><td>CN-</td></tr><tr><td>R</td><td>1</td><td>1</td><td>1</td></tr><tr><td>I</td><td>0.5</td><td>0</td><td>0</td></tr><tr><td>C</td><td>- 10<sup>- 4.72</sup></td><td>+10<sup>- 4.72</sup></td><td>+10<sup>- 4.72</sup></td></tr><tr><td>E</td><td>0.5 - 10<sup>- 4.72</sup></td><td>10<sup>- 4.72</sup></td><td>10<sup>- 4.72</sup></td></tr></table> <p>Ka = (10<sup>- 4.72</sup>)<sup>2</sup> / 0.5 - 10<sup>- 4.72</sup><br/>OR<br/>Ka = (10<sup>- 4.72</sup>)<sup>2</sup> / 0.5 - 10<sup>- 4.72</sup><br/>But student must say 10<sup>- 4.72</sup> is negligible or &lt;5% cannot just be ignored<br/>Ka = 7.26 X 10<sup>-10</sup></p> |            | HCN]                   | H+   | CN-  | R          | 1    | 1                       | 1    | I      | 0.5        | 0    | 0                                                                                               | C | - 10 <sup>- 4.72</sup> | +10 <sup>- 4.72</sup> | +10 <sup>- 4.72</sup> | E | 0.5 - 10 <sup>- 4.72</sup> | 10 <sup>- 4.72</sup> | 10 <sup>- 4.72</sup> | Many girls did not explain the approximation as it is a weak acid. |
| Criteria:                                                                                                                                                                                                                         | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                       |                                                                                                 |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| <ul style="list-style-type: none"><li>Writes the Ka expression</li><li>Substitutes correct numbers in (does not need an ice table)</li><li>Must make correct statement about negligible change in concentration for HCN</li></ul> | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |                                                                                                 |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| <ul style="list-style-type: none"><li>Provides main steps of the calculation does not show all working.</li></ul>                                                                                                                 | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |                                                                                                 |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| <ul style="list-style-type: none"><li>Provides some relevant steps of the calculation</li></ul>                                                                                                                                   | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |                       |                                                                                                 |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
|                                                                                                                                                                                                                                   | HCN]                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | H+                    | CN-                                                                                             |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| R                                                                                                                                                                                                                                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | 1                     | 1                                                                                               |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| I                                                                                                                                                                                                                                 | 0.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 0                     | 0                                                                                               |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| C                                                                                                                                                                                                                                 | - 10 <sup>- 4.72</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | +10 <sup>- 4.72</sup> | +10 <sup>- 4.72</sup>                                                                           |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |
| E                                                                                                                                                                                                                                 | 0.5 - 10 <sup>- 4.72</sup>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 10 <sup>- 4.72</sup>  | 10 <sup>- 4.72</sup>                                                                            |                                                                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                      |      |                                                                                                 |              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                        |      |      |            |      |                         |      |        |            |      |                                                                                                 |   |                        |                       |                       |   |                            |                      |                      |                                                                    |

| 26<br>b)                                                                                                                                                                                                                                                                                       | <table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td><ul style="list-style-type: none"><li>Shows all working using Ka expression with correct substitutions to calculate H<sub>3</sub>O<sup>+</sup></li><li>to calculate correctly the concentration of [NaClO ]</li><li>Calculates the pH</li><li>Define buffer using a correct equation</li></ul></td><td>4</td></tr><tr><td><ul style="list-style-type: none"><li>3 of the above</li><li></li></ul></td><td>3</td></tr><tr><td><ul style="list-style-type: none"><li>2 of the above or two mistakes</li><li></li></ul></td><td>2</td></tr><tr><td><ul style="list-style-type: none"><li>Some relevant information/ calculation OR used some data.</li></ul></td><td>1</td></tr></table> | Criteria: | Marks | <ul style="list-style-type: none"><li>Shows all working using Ka expression with correct substitutions to calculate H<sub>3</sub>O<sup>+</sup></li><li>to calculate correctly the concentration of [NaClO ]</li><li>Calculates the pH</li><li>Define buffer using a correct equation</li></ul> | 4 | <ul style="list-style-type: none"><li>3 of the above</li><li></li></ul>                                                                                                                | 3                                                         | <ul style="list-style-type: none"><li>2 of the above or two mistakes</li><li></li></ul> | 2 | <ul style="list-style-type: none"><li>Some relevant information/ calculation OR used some data.</li></ul> | 1 | <p>CALCULATE (3)</p> $K_a = \frac{[\text{ClO}][\text{H}^+]}{[\text{HClO}]}$ $[\text{H}^+] = K_a \times [\text{HClO}] / [\text{ClO}]$<br>$n(\text{NaClO}) = 0.107 \text{ g} / (22.99+35.45+16) = 0.107 \text{ g} / (74.44)= 0.001437\text{..mol}$ $[\text{NaClO} ] = n/ V$ $0.001437\text{..mol} / 0.1\text{L} = 0.014374 \text{ M}$<br>$[\text{H}^+] = 3.0 \times 10^{-8} \times 0.010 \text{ mol L}^{-1} / 0.014374$ $2.08710171 \times 10^{-8}$ $\text{pH} = -\log [2.08710171 \times 10^{-8} = 7.68$ <p>EXPLAIN (1)</p> <p>Define buffer using a correct equation</p> | <p>Reasonably well done but major omission was showing detail in working. Concentration of Na Cl O was omitted t times. Two separate marks given. 3 and 1</p> <p>Many omitted to say why solution was a buffer.</p> <p>MANY girls did not read the question. It is hypochlorous acid so cant have an iodine in it!</p> |
|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------|-----------------------------------------------------------------------------------------|---|-----------------------------------------------------------------------------------------------------------|---|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Criteria:                                                                                                                                                                                                                                                                                      | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |       |                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                        |                                                           |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>Shows all working using Ka expression with correct substitutions to calculate H<sub>3</sub>O<sup>+</sup></li><li>to calculate correctly the concentration of [NaClO ]</li><li>Calculates the pH</li><li>Define buffer using a correct equation</li></ul> | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |       |                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                        |                                                           |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>3 of the above</li><li></li></ul>                                                                                                                                                                                                                        | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |       |                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                        |                                                           |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>2 of the above or two mistakes</li><li></li></ul>                                                                                                                                                                                                        | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |       |                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                        |                                                           |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>Some relevant information/ calculation OR used some data.</li></ul>                                                                                                                                                                                      | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |       |                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                        |                                                           |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |
| 27<br>a)                                                                                                                                                                                                                                                                                       | <table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td><ul style="list-style-type: none"><li>Identifies correct filter as yellow</li></ul></td><td>1</td></tr></table>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | Criteria: | Marks | <ul style="list-style-type: none"><li>Identifies correct filter as yellow</li></ul>                                                                                                                                                                                                            | 1 | <p>Must select yellow as most appropriate colour as yellow is directly opposite blue.</p><br><p>Must select blue as most appropriate colour as yellow is directly opposite yellow.</p> | <p>Very straightforward question – and very well done</p> |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |
| Criteria:                                                                                                                                                                                                                                                                                      | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |           |       |                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                        |                                                           |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |
| <ul style="list-style-type: none"><li>Identifies correct filter as yellow</li></ul>                                                                                                                                                                                                            | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             |           |       |                                                                                                                                                                                                                                                                                                |   |                                                                                                                                                                                        |                                                           |                                                                                         |   |                                                                                                           |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                        |

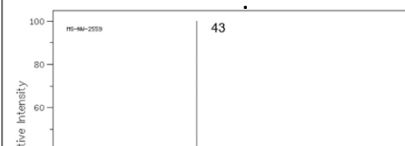
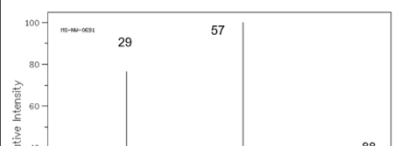
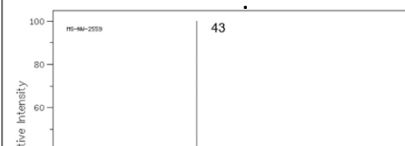
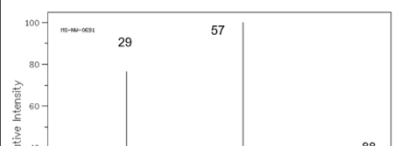
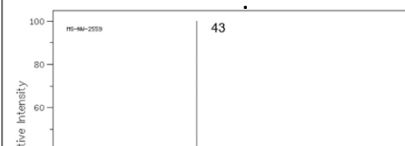
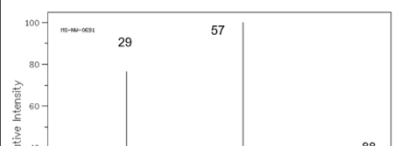
| 27<br>b)                                                                                                                                                                                                                                    | <table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td><ul style="list-style-type: none"><li>Labels axes correctly including units</li><li>Uses appropriate scale</li><li>Plots points correctly</li><li>Draws appropriate line of best fit with ruler</li></ul></td><td>3</td></tr><tr><td><ul style="list-style-type: none"><li>Provides a substantially correct graph</li></ul></td><td>2</td></tr><tr><td><ul style="list-style-type: none"><li>Includes one correct feature</li></ul></td><td>1</td></tr></table>                                 | Criteria: | Marks | <ul style="list-style-type: none"><li>Labels axes correctly including units</li><li>Uses appropriate scale</li><li>Plots points correctly</li><li>Draws appropriate line of best fit with ruler</li></ul>                                   | 3 | <ul style="list-style-type: none"><li>Provides a substantially correct graph</li></ul> | 2 | <ul style="list-style-type: none"><li>Includes one correct feature</li></ul>                | 1 |                                                                                                                                                                                                                                                                                                                                                               | Very well done across cohort                                                                                                                                                                                                                                                         |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------|---|---------------------------------------------------------------------------------------------|---|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Criteria:                                                                                                                                                                                                                                   | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>Labels axes correctly including units</li><li>Uses appropriate scale</li><li>Plots points correctly</li><li>Draws appropriate line of best fit with ruler</li></ul>                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>Provides a substantially correct graph</li></ul>                                                                                                                                                      | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>Includes one correct feature</li></ul>                                                                                                                                                                | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
| 27<br>c)                                                                                                                                                                                                                                    | <table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td><ul style="list-style-type: none"><li>Uses graph to find concentration at 0.444</li><li>Correct calculation of concentration based on graph</li><li>Correct statement comparing calculated value with EPA recommended guidelines.</li></ul></td><td>3</td></tr><tr><td><ul style="list-style-type: none"><li>Substantially correct</li></ul></td><td>2</td></tr><tr><td><ul style="list-style-type: none"><li>Includes one correct calculation or process</li></ul></td><td>1</td></tr></table> | Criteria: | Marks | <ul style="list-style-type: none"><li>Uses graph to find concentration at 0.444</li><li>Correct calculation of concentration based on graph</li><li>Correct statement comparing calculated value with EPA recommended guidelines.</li></ul> | 3 | <ul style="list-style-type: none"><li>Substantially correct</li></ul>                  | 2 | <ul style="list-style-type: none"><li>Includes one correct calculation or process</li></ul> | 1 | <p>From graph concentration is 0.7 mg/L</p> <p>So concentration of sample is 7 mg/L (was diluted by 10 for absorption)</p> <p>Which is 0.007g/L<br/>MM <math>\text{PO}_4^{3-} = 30.97 + (4 \times 16) = 94.97 \text{ g/mol}</math><br/><math>0.007\text{g/L} / 94.97 \text{ g/mol} = 7.37 \times 10^{-5}\text{mol/L}</math></p> <p>Does not comply as concentration is greater than EPA limit <math>5.0 \times 10^{-7} \text{ mol/L}</math></p> | <p>Ideally add the interpolation dotted line<br/>Many students struggled with the dilution</p> <p>Lots of answers don't include information on what is being calculated, should say <math>[\text{PO}_4^{3-}] = \dots</math></p> <p>Some students forgot to consider the dilution</p> |
| Criteria:                                                                                                                                                                                                                                   | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>Uses graph to find concentration at 0.444</li><li>Correct calculation of concentration based on graph</li><li>Correct statement comparing calculated value with EPA recommended guidelines.</li></ul> | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>Substantially correct</li></ul>                                                                                                                                                                       | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |
| <ul style="list-style-type: none"><li>Includes one correct calculation or process</li></ul>                                                                                                                                                 | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                                                                                                                                                             |   |                                                                                        |   |                                                                                             |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                      |

|                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <div><p style="text-align: center;">Absorbance</p><table border="1"><caption>Data points from the Absorbance vs Concentration graph</caption><thead><tr><th>Concentration (mg/L)</th><th>Absorbance</th></tr></thead><tbody><tr><td>0</td><td>0</td></tr><tr><td>0.4</td><td>0.05</td></tr><tr><td>1.2</td><td>0.15</td></tr><tr><td>2.0</td><td>0.25</td></tr><tr><td>3.6</td><td>0.48</td></tr></tbody></table></div> <p>From graph concentration is 2.55 mg/L<br/>Amount in 100 mL sample = 2.55 mg/L x 0.100 L = 0.255 mg</p> <p>0.0255 mg in 10 mL sample</p> <p><math>C = 0.0255 \text{ mg} / 0.010 \text{ L} = 2.55 \text{ mg/L} = 0. \text{g/L}</math><br/><math>\text{MM PO}_4^{3-} = 30.97 + (4 \times 16) = 94.97 \text{ g/mol}</math><br/><math>0.007 \text{ g/L} / 94.97 \text{ g/mol} = 7.37 \times 10^{-5} \text{ mol/L}</math></p> <p>Does not comply as concentration is greater than EPA limit <math>5.0 \times 10^{-7} \text{ mol/L}</math></p> | Concentration (mg/L) | Absorbance                                                                                                                 | 0 | 0                                                                                                                                      | 0.4 | 0.05                                                                | 1.2 | 0.15                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 2.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | 0.25 | 3.6 | 0.48 |  |
|----------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------|----------------------------------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------|-----|---------------------------------------------------------------------|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------|-----|------|--|
| Concentration (mg/L)                                                                                                                   | Absorbance                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| 0                                                                                                                                      | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| 0.4                                                                                                                                    | 0.05                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| 1.2                                                                                                                                    | 0.15                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| 2.0                                                                                                                                    | 0.25                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| 3.6                                                                                                                                    | 0.48                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| 27 d)                                                                                                                                  | <table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td><ul style="list-style-type: none"><li>Correct conclusion based on correct calculations</li><li>Must show working</li></ul></td><td>4</td></tr><tr><td><ul style="list-style-type: none"><li>Correct conclusion based on calculations (minor error) OR</li><li>Insufficient working</li></ul></td><td>3</td></tr><tr><td><ul style="list-style-type: none"><li>Provides main steps</li></ul></td><td>2</td></tr></table> | Criteria:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Marks                | <ul style="list-style-type: none"><li>Correct conclusion based on correct calculations</li><li>Must show working</li></ul> | 4 | <ul style="list-style-type: none"><li>Correct conclusion based on calculations (minor error) OR</li><li>Insufficient working</li></ul> | 3   | <ul style="list-style-type: none"><li>Provides main steps</li></ul> | 2   | <p><math>n(\text{PO}_4^{3-}) = 2.1 \times 10^{-4} \times 0.100 = 2.1 \times 10^{-5} \text{ moles}</math><br/><math>n(\text{Ca}^{2+}) = 5 \times 10^{-2} \times 0.200 = 0.01 \text{ moles}</math></p> <p>Total volume of solution 300 mL</p> <p><math>[\text{Ca}^{2+}] = 0.01 / 0.3 = 0.033333... \text{ mol/L}</math><br/><math>[\text{PO}_4^{3-}] = 2.1 \times 10^{-5} / 0.3 = 0.00007 \text{ mol/L}</math></p> <p><math>K_{sp} = [\text{Ca}^{2+}]^3 [\text{PO}_4^{3-}]^2</math></p> <p><math>Q = (0.033333...) ^3 \times (0.00007) ^2</math><br/><math>= 1.8148... \times 10^{-13}</math></p> | <p>Most students calculated Number of moles (1 mark) concentrations correctly (1 mark) – well done</p> <p>Lots of errors in formulating correct Ksp expression – resulting from not knowing charge on a phosphate ion(1 mark lost)</p> <p>Some students rounded correct concentration values in Ksp calculation (1 mark lost)</p> <p>Hence, Qsp was calculated incorrectly</p> <p>Most students demonstrated that if Qsp &gt; Ksp a precipitation will form (1 mark)</p> |      |     |      |  |
| Criteria:                                                                                                                              | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| <ul style="list-style-type: none"><li>Correct conclusion based on correct calculations</li><li>Must show working</li></ul>             | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| <ul style="list-style-type: none"><li>Correct conclusion based on calculations (minor error) OR</li><li>Insufficient working</li></ul> | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |
| <ul style="list-style-type: none"><li>Provides main steps</li></ul>                                                                    | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                      |                                                                                                                            |   |                                                                                                                                        |     |                                                                     |     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |      |     |      |  |

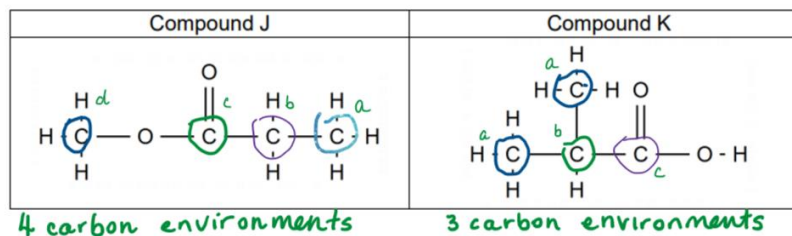
|                                                                                                 | <table><tr><td></td><td></td></tr><tr><td><ul style="list-style-type: none"><li>Provides some relevant steps of the calculation</li></ul></td><td>1</td></tr></table>                                                                                                                                                                                                                                                                                 |           |       | <ul style="list-style-type: none"><li>Provides some relevant steps of the calculation</li></ul> | 1 | <p>Since this product is greater than <math>K_{sp}</math> (<math>2.07 \times 10^{-29}</math>), a precipitate will form.</p> <p><math>n(\text{PO}_4^{3-}) = 5.0 \times 10^{-3} \times 0.100 = 5.0 \times 10^{-4}</math> moles<br/><math>n(\text{Cu}^{2+}) = 1.9 \times 10^{-5} \times 0.200 = 3.8 \times 10^{-6}</math> moles</p> <p>Total volume of solution 300 mL</p> <p><math>[\text{Cu}^{2+}] = 3.8 \times 10^{-6} / 0.3 = 1.266666 \times 10^{-5}</math> mol/L<br/><math>[\text{PO}_4^{3-}] = 5.0 \times 10^{-4} / 0.3 = 1.66666... \times 10^{-3}</math> mol/L</p> <p><math>K_{sp} = [\text{Cu}^{2+}]^3 [\text{PO}_4^{3-}]^2</math></p> <p><math>Q = (1.266666 \times 10^{-5})^3 \times (1.66666... \times 10^{-3})^2</math><br/><math>= 5.6 \times 10^{-21}</math></p> <p>Since this product is greater than <math>K_{sp}</math> (<math>1.4 \times 10^{-37}</math>), a precipitate will form.</p> |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
|-------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------|-------|-------------------------------------------------------------------------------------------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---|----|--|-------------|--|---------------|---|----|--|-------------------------------------------------------------------|--|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| <ul style="list-style-type: none"><li>Provides some relevant steps of the calculation</li></ul> | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| 28                                                                                              | <table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td>All 5 drawings are correct with ALL bonds shown</td><td>5</td></tr><tr><td>Structures correct but not all bonds shown e.g OH</td><td>4</td></tr><tr><td>OR</td><td></td></tr><tr><td>One mistake</td><td></td></tr><tr><td>Two mistakes'</td><td>3</td></tr><tr><td>OR</td><td></td></tr><tr><td>One mistake and structures correct but not all bonds shown e.g OH</td><td></td></tr></table> | Criteria: | Marks | All 5 drawings are correct with ALL bonds shown                                                 | 5 | Structures correct but not all bonds shown e.g OH                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 4 | OR |  | One mistake |  | Two mistakes' | 3 | OR |  | One mistake and structures correct but not all bonds shown e.g OH |  | <p>A – propan-1-ol</p> <pre>      H   H   H   O                     H — C — C — C — O                       H   H   H</pre> <p>B- propan-2-ol</p> <pre>      H       O                 H — C — C — C — H                       H   H   H</pre> <p>C propanoic acid</p> <pre>      H   H   O                  H — C — C — C — O — H                   H   H</pre> <p>D propanone</p> | <p>C – OH</p> <p>Please draw O-H with a bent bond (relates understanding of properties)</p> <p>Note that the propanamide would only occur with heat – no heat was shown, so that better answer is the acid/base</p> |
| Criteria:                                                                                       | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                 |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| All 5 drawings are correct with ALL bonds shown                                                 | 5                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| Structures correct but not all bonds shown e.g OH                                               | 4                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| OR                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| One mistake                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| Two mistakes'                                                                                   | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                     |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| OR                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |
| One mistake and structures correct but not all bonds shown e.g OH                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                       |           |       |                                                                                                 |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |   |    |  |             |  |               |   |    |  |                                                                   |  |                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                     |

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|                                                                                                                                                                                                                         | Three mistakes<br><br>OR<br><br>Two mistake and structures correct but not all bonds shown<br>e.g OH                                                                                                                                                                                                        | 2               | <div></div><br>E<br>ammonium propanoate (acid base reaction)                                                     |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
|                                                                                                                                                                                                                         | Four mistakes<br><br>OR<br><br>three mistake and structures correct but not all bonds shown<br>e.g OH                                                                                                                                                                                                       | 1               | <div></div><br>propenamide (propanoic acid is heated and the ammonium salt dehydrates producing amide and water) |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| 29<br>a)                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                             |                 |                                                                                                                                                                                                    | Well done. A few girls forgot the 2 litres. Please be careful of your maths in calculations |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
|                                                                                                                                                                                                                         | <table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td><ul style="list-style-type: none"><li>Writes the Keq expression<br/>OR</li><li>Substitutes correct numbers in to the Keq expressions</li><li>Uses ICE table and calculations from this to give correct answer</li></ul></td><td>3</td></tr></table> | Criteria:       | Marks                                                                                                                                                                                              |                                                                                             | <ul style="list-style-type: none"><li>Writes the Keq expression<br/>OR</li><li>Substitutes correct numbers in to the Keq expressions</li><li>Uses ICE table and calculations from this to give correct answer</li></ul> | 3 | <table><tr><td>M</td><td>O<sub>2</sub></td><td>SO<sub>2</sub></td><td>SO<sub>3</sub></td></tr><tr><td>R</td><td>1</td><td>2</td><td>2</td></tr><tr><td>I</td><td>1.0</td><td>0.445</td><td>0</td></tr><tr><td>C</td><td>-0.20</td><td>-0.40</td><td>+0.40</td></tr><tr><td>E</td><td>0.80</td><td>0.045</td><td>0.40</td></tr></table><br>OR<br><table><tr><td>mol</td><td>O<sub>2</sub></td><td>SO<sub>2</sub></td><td>SO<sub>3</sub></td></tr><tr><td>R</td><td>1</td><td>2</td><td>2</td></tr><tr><td>I</td><td>2</td><td>0.89</td><td>0</td></tr></table> | M | O <sub>2</sub> | SO <sub>2</sub> | SO <sub>3</sub> | R | 1 | 2 | 2 | I | 1.0 | 0.445 | 0 | C | -0.20 | -0.40 | +0.40 | E | 0.80 | 0.045 | 0.40 | mol | O <sub>2</sub> | SO <sub>2</sub> | SO <sub>3</sub> | R | 1 | 2 | 2 | I | 2 | 0.89 |
| Criteria:                                                                                                                                                                                                               | Marks                                                                                                                                                                                                                                                                                                       |                 |                                                                                                                                                                                                    |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| <ul style="list-style-type: none"><li>Writes the Keq expression<br/>OR</li><li>Substitutes correct numbers in to the Keq expressions</li><li>Uses ICE table and calculations from this to give correct answer</li></ul> | 3                                                                                                                                                                                                                                                                                                           |                 |                                                                                                                                                                                                    |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| M                                                                                                                                                                                                                       | O <sub>2</sub>                                                                                                                                                                                                                                                                                              | SO <sub>2</sub> | SO <sub>3</sub>                                                                                                                                                                                    |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| R                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                           | 2               | 2                                                                                                                                                                                                  |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| I                                                                                                                                                                                                                       | 1.0                                                                                                                                                                                                                                                                                                         | 0.445           | 0                                                                                                                                                                                                  |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| C                                                                                                                                                                                                                       | -0.20                                                                                                                                                                                                                                                                                                       | -0.40           | +0.40                                                                                                                                                                                              |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| E                                                                                                                                                                                                                       | 0.80                                                                                                                                                                                                                                                                                                        | 0.045           | 0.40                                                                                                                                                                                               |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| mol                                                                                                                                                                                                                     | O <sub>2</sub>                                                                                                                                                                                                                                                                                              | SO <sub>2</sub> | SO <sub>3</sub>                                                                                                                                                                                    |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| R                                                                                                                                                                                                                       | 1                                                                                                                                                                                                                                                                                                           | 2               | 2                                                                                                                                                                                                  |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |
| I                                                                                                                                                                                                                       | 2                                                                                                                                                                                                                                                                                                           | 0.89            | 0                                                                                                                                                                                                  |                                                                                             |                                                                                                                                                                                                                         |   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |   |                |                 |                 |   |   |   |   |   |     |       |   |   |       |       |       |   |      |       |      |     |                |                 |                 |   |   |   |   |   |   |      |

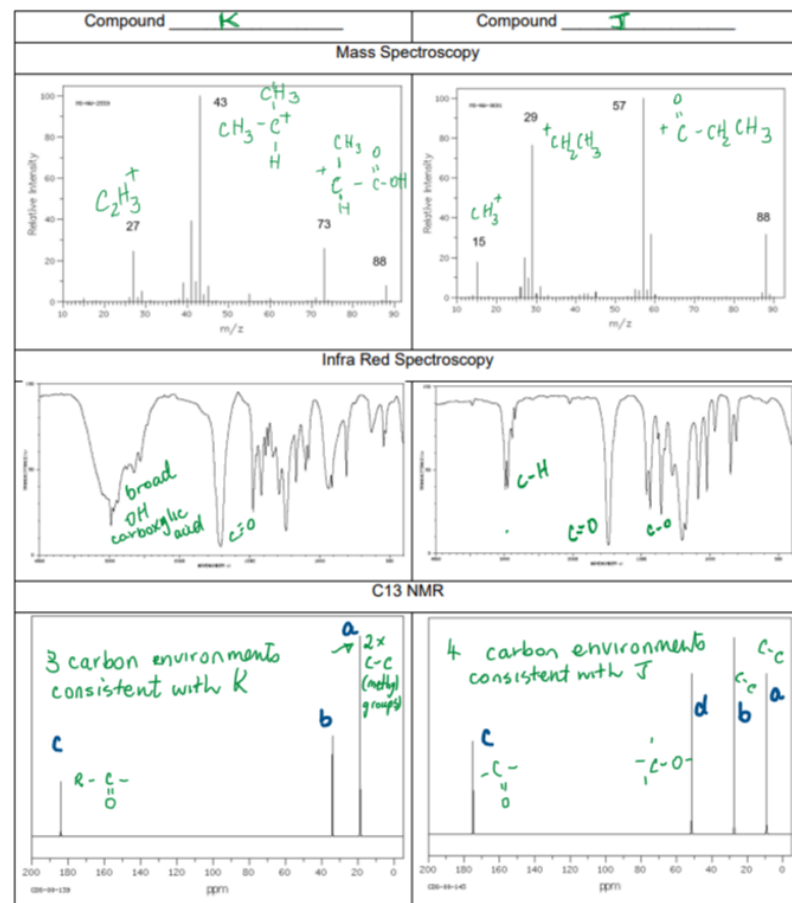
|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |            |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |     |      |     |           |     |       |     |  |
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|                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <ul style="list-style-type: none"><li></li><li>Provides main steps of the calculation does not show all working.</li><li>Provides some relevant steps of the calculation e.g writes the correct Keq expression</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2<br><br>1 | <table><tr><td>C</td><td>-0.4</td><td>-0.8</td><td>+0.8</td></tr><tr><td>E</td><td>1.6</td><td>0.09</td><td>0.8</td></tr><tr><td>[ ] = E/2</td><td>0.8</td><td>0.045</td><td>0.4</td></tr></table> <p>Keq = [SO<sub>3</sub>]<sup>2</sup>/([SO<sub>2</sub>]<sup>2</sup>[O<sub>2</sub>])<br/>= (0.40)<sup>2</sup> / ((0.045)<sup>2</sup>(0.8))<br/>= 98.77<br/>= 98.8</p> | C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -0.4 | -0.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | +0.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | E | 1.6 | 0.09 | 0.8 | [ ] = E/2 | 0.8 | 0.045 | 0.4 |  |
| C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | -0.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | -0.8       | +0.8                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |     |      |     |           |     |       |     |  |
| E                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | 1.6                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.09       | 0.8                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |     |      |     |           |     |       |     |  |
| [ ] = E/2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 0.8                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | 0.045      | 0.4                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |     |      |     |           |     |       |     |  |
| 29 b)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <table><tr><td>Criteria:</td><td>Marks</td></tr><tr><td><ul style="list-style-type: none"><li>Thorough analysis of temperature and pressure can include catalyst but not needed</li><li>Ratio- forces equilibrium to the right</li><li>At LEAST two conditions are necessary and analysed in detail</li><li>Essential to talk about compromise for at least one condition</li><li>Uses data given.</li></ul><p>Temperature:</p><ul style="list-style-type: none"><li>High temp related to ROR (collision)</li><li>Low temp related to LCP (yield)</li><li>Therefore compromise</li></ul><p>Pressure</p><ul style="list-style-type: none"><li>LCP high pressure (yield)</li></ul></td><td>3</td></tr></table> | Criteria:  | Marks                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>Thorough analysis of temperature and pressure can include catalyst but not needed</li><li>Ratio- forces equilibrium to the right</li><li>At LEAST two conditions are necessary and analysed in detail</li><li>Essential to talk about compromise for at least one condition</li><li>Uses data given.</li></ul> <p>Temperature:</p> <ul style="list-style-type: none"><li>High temp related to ROR (collision)</li><li>Low temp related to LCP (yield)</li><li>Therefore compromise</li></ul> <p>Pressure</p> <ul style="list-style-type: none"><li>LCP high pressure (yield)</li></ul> | 3    | <p>This is a compromising conditions question AND must relate yield and equilibrium.</p> <p><b>In industrial synthesis the aim is to maximise yield and reaction rate without compromising cost and safety .</b></p> <p><b>At least 2</b> of the following:</p> <p>Reactants are added in ratio 1:1 so excess oxygen is provided, reaction stoichiometry requires 1:2 ratio of O<sub>2</sub>:SO<sub>2</sub> this will drive equilibrium to the right. Oxygen is also easier and safer to recycle (toxicity of SO<sub>2</sub>) and so this is the compromise situation.</p> <p>High temperature of 400-450°C increases the energy of reacting particles and provides a greater chance for successful collisions hence increasing the rate. However as the reaction is exothermic the equilibrium would shift to the left so a compromise is needed. This temperature provides energy to reach Ea but a compromise exists between rate and equilibrium</p> <p>Increase in pressure to 200 KPa would drive the equilibrium to the right and increase yield. Using pressures too high can be costly and dangerous, so must be balance with benefit/ safety- again a compromise is necessary. .</p> <p>Using a V<sub>2</sub>O<sub>5</sub> catalyst lowers the activation energy for the reaction and increases the rate of reaction, however, yield is the same so the only way that it is an advantage is that it may become a less costly process.</p> | <p>There are 4 conditions mentioned so it is important that AT LEAST TWO are referred to in the analysis AND THAT THE COMPROMISE FOR AT LEAST ONE IS DISCUSSED.</p> <p>The question requires analysis of the conditions that are used, and you should be justifying these conditions and it is essential you talk about the compromise necessary. It is not just looking at the conditions. NOTE THE AIM OF INDUSTRIAL SYNTHESIS!.</p> <p>Many girls explained conditions but did not mention the need to compromise to get the best yield.</p> <p>Many girls thought that 400oC was a LOW temperature.</p> <p>Best answers mentioned the compromising situation in TWO factors, and referred to data.</p> <p>Many answers contained LOTS of unnecessary information.</p> <p>If you talk about catalyst, then you need to analyse why this is used and to what advantage. ( and state that it has no effect on yield so really isnt fulfilling the aim of industrial synthesis completely. )</p> |   |     |      |     |           |     |       |     |  |
| Criteria:                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |            |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |     |      |     |           |     |       |     |  |
| <ul style="list-style-type: none"><li>Thorough analysis of temperature and pressure can include catalyst but not needed</li><li>Ratio- forces equilibrium to the right</li><li>At LEAST two conditions are necessary and analysed in detail</li><li>Essential to talk about compromise for at least one condition</li><li>Uses data given.</li></ul> <p>Temperature:</p> <ul style="list-style-type: none"><li>High temp related to ROR (collision)</li><li>Low temp related to LCP (yield)</li><li>Therefore compromise</li></ul> <p>Pressure</p> <ul style="list-style-type: none"><li>LCP high pressure (yield)</li></ul> | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |            |                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |   |     |      |     |           |     |       |     |  |

|                                                                                      | <ul style="list-style-type: none"><li>Low pressure chosen for safety OR cost reasons</li><li></li></ul>                                                                                               |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
|--------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|-------|---------------------------------------------|---|--|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------|-------------------|-------------------|--|--------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|--|
|                                                                                      | <ul style="list-style-type: none"><li>Thorough analysis of temperature and/or pressure required</li><li>OR</li><li>TEMP/ PRESS less well analysed and no mention of compromise and catalyst</li></ul> | 2        |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
|                                                                                      | <ul style="list-style-type: none"><li>Analysis of 1 relevant factor which includes:</li><li>Specific reference to flowchart data</li><li>OR</li><li>Effect on rate and yield</li></ul>                | 1        |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
| 30 a                                                                                 | <table><tr><th>Criteria</th><th>Marks</th></tr><tr><td>Correctly identifies the type of isomerism.</td><td>1</td></tr></table>                                                                        | Criteria | Marks | Correctly identifies the type of isomerism. | 1 |  | Functional group isomers                                                                                                                                                                                                                                                                                                        |                   |                   |                   |  |                                                                                      |                                                                                       |  |
| Criteria                                                                             | Marks                                                                                                                                                                                                 |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
| Correctly identifies the type of isomerism.                                          | 1                                                                                                                                                                                                     |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
| 30 b                                                                                 | <table><tr><th>Criteria</th><th>marks</th></tr><tr><td></td><td>1</td></tr></table>                                                                                                                   | Criteria | marks |                                             | 1 |  | <table><tr><td>Compound <b>K</b></td><td>Compound <b>J</b></td></tr><tr><td colspan="2">Mass Spectroscopy</td></tr><tr><td></td><td></td></tr></table> | Compound <b>K</b> | Compound <b>J</b> | Mass Spectroscopy |  |  |  |  |
| Criteria                                                                             | marks                                                                                                                                                                                                 |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
|                                                                                      | 1                                                                                                                                                                                                     |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
| Compound <b>K</b>                                                                    | Compound <b>J</b>                                                                                                                                                                                     |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
| Mass Spectroscopy                                                                    |                                                                                                                                                                                                       |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |
|  |                                                                                                                  |          |       |                                             |   |  |                                                                                                                                                                                                                                                                                                                                 |                   |                   |                   |  |                                                                                      |                                                                                       |  |

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------------------|------------------------------------------------|------------------------------------------------|--------------------------------------|------|----|----|----|----|----|-----------------------------------|------------------|----------------------------------|-----------------------------------------------|-----------------------------------------------|-----------------------------------------------|---------------------------------|
| 30                                                                                                                                                                                                                          | Criteria:                                                                                                                                                                                                                     | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                      | For methyl propanoic acid<br>markers reference<br>Methyl propanoic acid                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <p>Student should not just be doing process of elimination but using positive results/ reasons to why they made their choice. Students should be drawing/ labelling the environments if referring to them</p> <p>Really good attempt by all just lacking a thorough explanation for ALL SPECTRA DATA – this includes explaining all relevant peaks- For the majority of students CNMR peaks were ignored and only the number of environments were addressed – you have this information on the data sheet so should be referring/ using it. Although this was a simple spectra to work out which compound belonged to which spectra – it was a 6 mark questions with 6 spectra to be discussed. 6 marks you should spend minimum 10 mins. - detail was lacking to attain 5/6 and 6/6 marks for many – would not have spent 10 mins on this. Lots of students not including cation + charge on the fragmentations or putting – charges OR free radicals. Many students calling the –OH from the acid an alcohol group. Also, should be VERY BROAD PEAK. Cannot just say broad peak at 3000 is an acid – it represents an OH group present within an acid. Note for students – the ‘dips’ in IR spec are still referred to as peaks</p> <p>Poorly done/ absent<br/>WITH CORRECT CATION CHARGE</p> |                                                |                                                |                                               |                                               |                                               |                                               |                                               |                                               |    |    |    |                                   |                                                              |                                                |                     |                                 |                                               |                                               |                                               |                                               |                                               |                                               |                                      |    |    |      |    |    |    |                                   |                                                              |                                                              |                                                |                                                |                                                |                                                |                                      |      |    |    |    |    |    |                                   |                  |                                  |                                               |                                               |                                               |                                 |
|                                                                                                                                                                                                                             | <ul style="list-style-type: none"><li>Justifies the correct structure showing an extensive understanding of the interpretation of spectroscopic data</li><li>Refers explicitly to all 3 spectroscopic data diagrams</li></ul> | 6                                                                                                                                                                                                                                                                                                                                                                                                                                                          | <table><tr><td>m/z value of [fragment]<sup>+</sup></td><td>73</td><td>55</td><td>45</td><td>44 ?</td><td>43</td><td>42</td><td>41</td><td>39</td><td>29</td><td>27</td></tr><tr><td>[molecular fragment]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>5</sub>O<sub>2</sub>]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>3</sub>O]<sup>+</sup></td><td>[COOH]<sup>+</sup></td><td>[CO<sub>2</sub>]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>7</sub>]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>6</sub>]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>5</sub>]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>3</sub>]<sup>+</sup></td><td>[C<sub>2</sub>H<sub>5</sub>]<sup>+</sup></td><td>[C<sub>2</sub>H<sub>3</sub>]<sup>+</sup></td></tr></table><br><table><tr><td>m/z value of [fragment]<sup>+</sup></td><td>87</td><td>59</td><td>58 ?</td><td>57</td><td>56</td><td>55</td></tr><tr><td>[molecular fragment]<sup>+</sup></td><td>[C<sub>4</sub>H<sub>6</sub>O<sub>2</sub>]<sup>+</sup></td><td>[C<sub>2</sub>H<sub>3</sub>O<sub>2</sub>]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>6</sub>O]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>5</sub>O]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>4</sub>O]<sup>+</sup></td><td>[C<sub>3</sub>H<sub>3</sub>O]<sup>+</sup></td></tr></table><br><table><tr><td>m/z value of [fragment]<sup>+</sup></td><td>45 ?</td><td>31</td><td>29</td><td>27</td><td>26</td><td>15</td></tr><tr><td>[molecular fragment]<sup>+</sup></td><td>[?]<sup>+</sup></td><td>[CH<sub>3</sub>O]<sup>+</sup></td><td>[C<sub>2</sub>H<sub>5</sub>]<sup>+</sup></td><td>[C<sub>2</sub>H<sub>3</sub>]<sup>+</sup></td><td>[C<sub>2</sub>H<sub>2</sub>]<sup>+</sup></td><td>[CH<sub>3</sub>]<sup>+</sup></td></tr></table> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | m/z value of [fragment] <sup>+</sup>           | 73                                             | 55                                            | 45                                            | 44 ?                                          | 43                                            | 42                                            | 41                                            | 39 | 29 | 27 | [molecular fragment] <sup>+</sup> | [C <sub>3</sub> H <sub>5</sub> O <sub>2</sub> ] <sup>+</sup> | [C <sub>3</sub> H <sub>3</sub> O] <sup>+</sup> | [COOH] <sup>+</sup> | [CO <sub>2</sub> ] <sup>+</sup> | [C <sub>3</sub> H <sub>7</sub> ] <sup>+</sup> | [C <sub>3</sub> H <sub>6</sub> ] <sup>+</sup> | [C <sub>3</sub> H <sub>5</sub> ] <sup>+</sup> | [C <sub>3</sub> H <sub>3</sub> ] <sup>+</sup> | [C <sub>2</sub> H <sub>5</sub> ] <sup>+</sup> | [C <sub>2</sub> H <sub>3</sub> ] <sup>+</sup> | m/z value of [fragment] <sup>+</sup> | 87 | 59 | 58 ? | 57 | 56 | 55 | [molecular fragment] <sup>+</sup> | [C <sub>4</sub> H <sub>6</sub> O <sub>2</sub> ] <sup>+</sup> | [C <sub>2</sub> H <sub>3</sub> O <sub>2</sub> ] <sup>+</sup> | [C <sub>3</sub> H <sub>6</sub> O] <sup>+</sup> | [C <sub>3</sub> H <sub>5</sub> O] <sup>+</sup> | [C <sub>3</sub> H <sub>4</sub> O] <sup>+</sup> | [C <sub>3</sub> H <sub>3</sub> O] <sup>+</sup> | m/z value of [fragment] <sup>+</sup> | 45 ? | 31 | 29 | 27 | 26 | 15 | [molecular fragment] <sup>+</sup> | [?] <sup>+</sup> | [CH <sub>3</sub> O] <sup>+</sup> | [C <sub>2</sub> H <sub>5</sub> ] <sup>+</sup> | [C <sub>2</sub> H <sub>3</sub> ] <sup>+</sup> | [C <sub>2</sub> H <sub>2</sub> ] <sup>+</sup> | [CH <sub>3</sub> ] <sup>+</sup> |
|                                                                                                                                                                                                                             | m/z value of [fragment] <sup>+</sup>                                                                                                                                                                                          | 73                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 55                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 45                                             | 44 ?                                           | 43                                            | 42                                            | 41                                            | 39                                            | 29                                            | 27                                            |    |    |    |                                   |                                                              |                                                |                     |                                 |                                               |                                               |                                               |                                               |                                               |                                               |                                      |    |    |      |    |    |    |                                   |                                                              |                                                              |                                                |                                                |                                                |                                                |                                      |      |    |    |    |    |    |                                   |                  |                                  |                                               |                                               |                                               |                                 |
|                                                                                                                                                                                                                             | [molecular fragment] <sup>+</sup>                                                                                                                                                                                             | [C <sub>3</sub> H <sub>5</sub> O <sub>2</sub> ] <sup>+</sup>                                                                                                                                                                                                                                                                                                                                                                                               | [C <sub>3</sub> H <sub>3</sub> O] <sup>+</sup>                                                              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<sub>3</sub> H <sub>5</sub> ] <sup>+</sup> | [C <sub>3</sub> H <sub>3</sub> ] <sup>+</sup> | [C <sub>2</sub> H <sub>5</sub> ] <sup>+</sup> | [C <sub>2</sub> H <sub>3</sub> ] <sup>+</sup> |    |    |    |                                   |                                                              |                                                |                     |                                 |                                               |                                               |                                               |                                               |                                               |                                               |                                      |    |    |      |    |    |    |                                   |                                                              |                                                              |                                                |                        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|                                                                                                                                                                                                                             | m/z value of [fragment] <sup>+</sup>                                                                                                                                                                                          | 87                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 59                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 58 ?                                           | 57                                             | 56                                            | 55                                            |                                               |                                               |                                               |                                               |    |    |    |                                   |                                                              |                                                |                     |                                 |                                               |                                               |                                               |                                               |                                               |                                               |                                      |    |    |      |    |    |    |                                   |                                                              |                                                              |                                                |                                                |                                                |                                                |                                      |      |    |    |    |    |    |                                   |                  |                                  |                                               |                                               |                                               |                                 |
| [molecular fragment] <sup>+</sup>                                                                                                                                                                                           | [C <sub>4</sub> H <sub>6</sub> O <sub>2</sub> ] <sup>+</sup>                                                                                                                                                                  | [C <sub>2</sub> H <sub>3</sub> O <sub>2</sub> ] <sup>+</sup>                                                                                                                                                                                                                                                                                                                                                                                               | [C <sub>3</sub> H <sub>6</sub> O] <sup>+</sup>                                                              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                                |                                               |                                               |                                               |    |    |    |                                   |                                                              |                                                |                     |                                 |                                               |                                               |                                               |                                               |                                               |                                               |                                      |    |    |      |    |    |    |                                   |                                                              |                                                              |                                                |                                   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| m/z value of [fragment] <sup>+</sup>                                                                                                                                                                                        | 45 ?                                                                                                                                                                                                                          | 31                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 29                                                                                                          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| [molecular fragment] <sup>+</sup>                                                                                                                                                                                           | [?] <sup>+</sup>                                                                                                                                                                                                              | [CH <sub>3</sub> O] <sup>+</sup>                                                                                                                                                                                                                                                                                                                                                                                                                           | [C <sub>2</sub> H <sub>5</sub> ] <sup>+</sup>                                                               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                                |                                               |                                               |                                               |    |    |    |                                   |                                                              |                                                |                     |                                 |                                               |                                               |                                               |                                               |                                               |                                               |                                      |    |    |      |    |    |    |                                   |                                                              |                                                              |                                                |                                                |                                                |                                                |                                      |      |    |    |    |    |    |                                   |                  |                                  |                                               |                                               |                                               |                                 |
| <ul style="list-style-type: none"><li>Justifies the structure showing a thorough understanding of the interpretation of spectroscopic data</li><li>Refers to relevant spectroscopic data in 3 diagrams</li></ul>            | 5                                                                                                                                                                                                                             | <p>Methyl propanoate</p> <p>3/6 was usually when there was a brief outline of each spectrum</p> <p>4/6 was usually when there was a brief outline of each spectrum with a little more detail for CNMR or MS</p> <p>5/6 was usually when there was comprehensive outline of each spectrum – explaining all peaks in each spectrum perhaps missing something like charges on the fragmentations of not referring to the CNMR peaks.</p>                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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             |                                                |                                                |                                      |      |    |    |    |    |    |                                   |                  |                                  |                                               |                                               |                                               |                                 |
| <ul style="list-style-type: none"><li>Shows a good understanding of the interpretation of spectroscopic data</li><li>Uses relevant information presented in the question to justify the structure of the compound</li></ul> | 4                                                                                                                                                                                                                             | <p>6/6 - To attain a thorough the students must have completed the following as a minimum.</p> <p>IR – 3000 broad peak = OH from an acid / absence from the other spectrum; as a narrow peak indicative of a C-H was at 3000.</p> <p>1700 – C=O which was in both</p>                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      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             |                                                |                                                |                                      |      |    |    |    |    |    |                                   |                  |                                  |                                               |                                               |                                               |                                 |
| <ul style="list-style-type: none"><li>Demonstrates some understanding of the interpretation of spectroscopic data</li><li>Provides some relevant information</li></ul>                                                      | 2-3<br><br>1                                                                                                                                                                                                                  | <p>MS – needn’t discuss the parent peak as already provided that they are isomers. Must have one unique peak per spectrum for each molecule identified what the fragment would be – WITH CORRECT CATION CHARGE</p> <p>CNMR- labelled diagrams identifying the carbon environments. Idea of symmetry to only give 3 peaks when there are 4 carbons</p> <p>Each peak identified on the CNMR with relation to the data sheet or spins / electronegativity</p> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |                                                |                                                |                                               |                                               |                                               |                                               |                                               |                                               |    |    |    |                                   |                                                              |                                                |                     |                                 |                                               |                                               |                                               |                                               |                                               |                                               |                                      |    |    |      |    |    |    |                                   |                                                              |                                                              |                                                |                                                |                                                |                                                |                                      |      |    |    |    |    |    |                                   |                  |                                  |                                               |                                               |                                               |                                 |



Each peak identified on the CNMR with relation to the data sheet or spins / electronegativity



| <div>31<br/>a)</div>                                                                                                                                                                                                                                                | <table><tr><th>Criteria:</th><th>Marks</th></tr><tr><td><ul style="list-style-type: none"><li>Calculating moles of both reagents. With working –</li><li>Comparing moles in the correct ratio to determine the limiting reagent</li></ul></td><td>2</td></tr><tr><td><ul style="list-style-type: none"><li>Calculated the moles of both reagents correctly but failed to compare moles in correct ration OR made a mistake with one of the mole calculations but has compared in the ratio correctly with the incorrect moles.</li></ul></td><td>1</td></tr></table> | Criteria:                                                                                                                                                                                                                                                                    | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | <ul style="list-style-type: none"><li>Calculating moles of both reagents. With working –</li><li>Comparing moles in the correct ratio to determine the limiting reagent</li></ul> | 2 | <ul style="list-style-type: none"><li>Calculated the moles of both reagents correctly but failed to compare moles in correct ration OR made a mistake with one of the mole calculations but has compared in the ratio correctly with the incorrect moles.</li></ul> | 1 | <p>2KOH + H<sub>2</sub>SO<sub>4</sub> --&gt; K<sub>2</sub>SO<sub>4</sub> + 2H<sub>2</sub>O<br/>n(KOH) = 5/ (39.1+16+1.008) = 0.08911385185 mol<br/>n(H<sub>2</sub>SO4) = CV. 0.5 x 0.08 = 0.04mol<br/>KOH : H<sub>2</sub>SO<sub>4</sub><br/>2:1<br/>0.08 : 0.04<br/>0.08 &lt; 0.08911.. therefore KOH is in excess and H<sub>2</sub>SO<sub>4</sub> is LR.<br/>Calculating moles of both reagents. With working<br/>0.08911385185 mol?</p> | <p>Mostly answered well - most students were able to calculate the moles of both reactants – there were a number of students that did not calculate the moles for both – the clue of Limiting reagent was even give to students. There needed to be a clear mole ratio step<br/>Many students simply calculating the moles and then H2SO4 is the LR – without the mole ratio.</p> <p>Stating the ratio clearly - examples<br/>KOH : H<sub>2</sub>SO<sub>4</sub><br/>2:1<br/>0.08911385185 &gt; 2 x 0.04 YES<br/>OR<br/>0.08911385185 &gt; 0.04 NO</p> |
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| Criteria:                                                                                                                                                                                                                                                           | Marks                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"><li>Calculating moles of both reagents. With working –</li><li>Comparing moles in the correct ratio to determine the limiting reagent</li></ul>                                                                                   | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <ul style="list-style-type: none"><li>Calculated the moles of both reagents correctly but failed to compare moles in correct ration OR made a mistake with one of the mole calculations but has compared in the ratio correctly with the incorrect moles.</li></ul> | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| <div>31<br/>b)</div>                                                                                                                                                                                                                                                |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <p>moles of H2O = 0.08 (from previous question or error carried forward)<br/>ΔH = -59.1 X 0.08 = -4.728 kJ = -4728J<br/>ΔH = -57.6 X 0.08911 = -5.132736 = -5132.736J<br/>ΔH = - MC ΔT<br/>ΔT = (-4728+5132.736)/ 80 X 4.18 = 29.48785 K</p> <p>29.48785 + 24.3 = 53.8°C</p> | <p>Students really struggled with the question – which is rightly so: it was hard and the last maths question of the paper, so you expect it to be hard. 1/3 marks should have been accessible for all students<br/>2/3 only a very small error<br/>3/3 PERFECT – band 6 candidate</p> <p>Note this question has been written in past examinations with an additional need to state any assumptions that were made during the calculation:<br/>Assuming....<br/>1. the solution has the same heat capacity as water</p> |                                                                                                                                                                                   |   |                                                                                                                                                                                                                                                                     |   |                                                                                                                                                                                                                                                                                                                                                                                                                                           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |

|    |                                                                                                                                                                                                                                                                                               |       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|    | Criteria:                                                                                                                                                                                                                                                                                     | Marks |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | <ol style="list-style-type: none"> <li>the combined mass of water is 80g from the solution.</li> <li>same density of water 1g=1mL</li> <li>No heat loss to the surroundings or cup</li> <li>the temperature increase will be dependent on the amount of energy released only</li> </ol>                                                                                                                                                                                                      |
|    | <ul style="list-style-type: none"> <li>Shows working of both <math>\Delta H</math>s</li> <li>Substitutes correct number for <math>\Delta H = - MC \Delta T</math></li> <li>Adds change <math>\Delta T</math> to initial temperature to correctly calculate the maximum temperature</li> </ul> | 3     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|    | <ul style="list-style-type: none"> <li>Calculates a q value for one equation</li> </ul> <p>OR Missing one step from above or incorrectly completes one step above – all other steps correct</p>                                                                                               | 2     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|    | <ul style="list-style-type: none"> <li>Provides some relevant calculations</li> </ul>                                                                                                                                                                                                         | 1     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 32 | Criteria:                                                                                                                                                                                                                                                                                     | Marks | <p><b>Biofuels</b> are derived from organic materials like plants and waste products.</p> <p>Biodiesel is a renewable fuel made from vegetable oils or animal fats..</p> <p>Ethanol is a biofuel made from renewable sources such as corn or sugarcane using the fermentation process:</p> $C_6H_{12}O_6 \rightarrow CO_2 + C_2H_5OH$ <p>(yeast is used as the catalyst)<br/>(temperature 40 °C)</p> <p>They undergo combustion reactions to release energy in the form of heat and produce carbon dioxide and water as byproducts.</p> | <p>This was marked holistically on how well you developed your argument.</p> <p>Students needed to build their argument using the data ie formulae and heats of combustion with extensive reference to chemical equations to gain full marks.</p> <p>Students also supported their answer with discussion of</p> <ul style="list-style-type: none"> <li>Environmental/emissions/pollution: (Acidification of oceans, greenhouse gas, Incomplete combustion Impurities of sulphur)</li> </ul> |
|    | <ul style="list-style-type: none"> <li>Judgement statement</li> <li>Thorough reference to data and links to argument</li> <li>Extensive use of appropriate chemical equations</li> </ul>                                                                                                      | 6     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
|    | <ul style="list-style-type: none"> <li>Judgement statement</li> <li>Refers to data and links to argument</li> <li>Good use of appropriate chemical equations</li> </ul>                                                                                                                       | 4-5   |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |

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| <ul style="list-style-type: none"> <li>General statement</li> <li>Some Reference to data and links to argument</li> <li>Minimal use of appropriate chemical equations</li> </ul> | 3 | <p>For example biodiesel</p> $2\text{C}_{11}\text{H}_{26}\text{O}_2 + 17\text{O}_2 \rightarrow 22\text{CO}_2 + 26\text{H}_2\text{O}$ <p>This releases 40-48 kJ/g of heat which is comparable to diesel, both have long carbon chains. However, biodiesel has the advantage of containing oxygen which promotes complete combustion. This minimises production of carbon which can irritate the lungs and carbon monoxide which is toxic to humans.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <ul style="list-style-type: none"> <li>Economic &amp; Energy efficiency</li> <li>Land use and</li> <li>Infrastructure issues</li> <li>Photosynthesis/ carbon neutral discussion</li> <li>Fermentation of glucose to produce bioethanol</li> </ul> <p>Only a handful of students indicated how biodiesel is made – this would have helped your argument supporting biofuels.</p> |
|                                                                                                                                                                                  | 2 | <p>Ethanol has a lower energy content (29.78 kJ/g) compared to octane (48 kJ/g) or diesel (44 kJ/g), but it burns more cleanly, producing fewer pollutants. The disadvantage is needing more volume to produce the same energy output. However, being miscible with traditional petrol (mainly octane), can be used as an additive, and reduce the amount of fossil fuel used.</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |                                                                                                                                                                                                                                                                                                                                                                                 |
|                                                                                                                                                                                  | 1 | <p><math>\text{C}_2\text{H}_5\text{OH} + \text{O}_2 \rightarrow 2\text{CO}_2 + 3\text{H}_2\text{O}</math></p> <p>The carbon dioxide released during biofuel combustion is part of the natural carbon cycle. Plants absorb carbon dioxide during their growth, and when biofuels are burned, the <math>\text{CO}_2</math> released is reabsorbed by new plants, making biofuels potentially carbon-neutral. Chemical equation for photosynthesis:</p> $6\text{CO}_2 + 6\text{H}_2\text{O} \rightarrow \text{C}_6\text{H}_{12}\text{O}_6 + 6\text{O}_2$ <p><b>Combustion of Fossil Fuels:</b> Fossil fuels, such as coal, oil, and natural gas, are derived from ancient organic matter that has undergone geological processes over millions of years. When fossil fuels are burned, they also undergo a combustion reaction:</p> <p>e.g. Octane</p> $2\text{C}_8\text{H}_{18} + 25\text{O}_2 \rightarrow 16\text{CO}_2 + 18\text{H}_2\text{O}$ <p>This releases 48 kJ/g of heat</p> <p>Being a long carbon chain may undergo incomplete combustion</p> $2\text{C}_8\text{H}_{18} + 24\text{O}_2 \rightarrow 14\text{CO}_2 + 2\text{CO} + 2\text{C} + 18\text{H}_2\text{O}$ <p>e.g. Diesel</p> $2\text{C}_{12}\text{H}_{26} + 37\text{O}_2 \rightarrow 24\text{CO}_2 + 26\text{H}_2\text{O}$ |                                                                                                                                                                                                                                                                                                                                                                                 |

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|  |  | <p>Diesel releases more CO<sub>2</sub> than biodiesel, and unlike biofuels, fossil fuels release carbon dioxide that has been sequestered in the Earth's crust for millions of years. This contributes to an increase in atmospheric CO<sub>2</sub> levels, which is a major factor in climate change. Also supplies of fossil fuels are limited and may run out in the future</p> <p>Both biodiesel, diesel and octane release a similar amount of heat energy that can be utilised.</p> <p>Judgement: Biofuels are a better option than fossil fuels as an energy source.</p> |  |
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